CivicConnect Project Engineering Document

PED v4.0 Final | M4 project closure and final evidence record

This document closes the CivicConnect project against the approved M1 requirements and the integrated M3 application. It records final scope status, acceptance evidence, unresolved limitations, the deployment decision and the project conclusion. Sections 45 onward provide the authoritative M4 status and supersede earlier milestone snapshots where they differ.

The application and evidence baseline assessed for this closure is GitHub main commit c8fcc9887d58bc07e7b592e292dbe05222d97190. The requirement-level scope calculation is 16.0 of 20 signed-off M1 requirement rows (80%). The team recommends no-go for production release because the integrated website load profile failed the request failure-rate threshold. The local academic demonstration may proceed only as a controlled, synthetic-data demonstration. The team verified that acceptance evidence for NFR-01 is absent from the assessed baseline; NFR-01 therefore remains unmet as a Must Have requirement.

Project Team and Collective Responsibility

The four registered students form the CivicConnect project team. The team shares responsibility for the PED, product, evidence and presentation.

Registered team member	**Name**	**Student number**	**Shared responsibility**
1	Marco Brazao de Sousa	601587	Collective team responsibility
2	Gift Mathe	600731	Collective team responsibility
3	Masana Sakala	600688	Collective team responsibility
4	Jacobus Petrus Geldenhuys	602755	Collective team responsibility

Problem, Stakeholders & Scope

1. Problem & Business Need

2. Identify stakeholders

3. Stakeholder needs

4. Stakeholder influence/interest

5. Stakeholder conflicts/competing expectations

6. In-scope items

7. Out-of-scope items

8. Deferred/future scope

9. Justify at least one exclusion/deferment

Requirements and RTM

1. Functional Requirements (FRs)

2. Non-Functional Requirements (NFRs)

3. Prioritise requirements

4. Give every requirement a unique ID

5. Identify requirement source/stakeholder

6. Write measurable acceptance criteria

7. Create initial Requirements Traceability Matrix (RTM)

8. Demonstrate at least one requirement traced from stakeholder/source → requirement → acceptance criteria

Constraints Risks and Forward Engineering

1. Scope constraints

2. Schedule constraints

3. Cost/resource constraints

4. Quality constraints

5. Security constraints

6. Explain interactions/trade-offs between constraints

7. Create Risk Register

8. Include probability, impact, priority, mitigation, contingency, owner and status

9. Identify 5–7 Forward Engineering Considerations

10. Explain why each matters now and the risk of ignoring it

Decisions GitHub and Project Control

1. Create Engineering Decision Log

2. Record decisions already made

3. Record deliberately deferred decisions

4. Set up/manage GitHub repository

5. GitHub issues/tasks

6. Branches and meaningful commits

7. Pull Requests/reviews/approvals

8. Protected main branch/governance

9. Create and maintain AI Usage Register

10. PED v1.0 version/document control

11. Coordinate team review

12. Baseline sign-off/gate evidence

1. Problem / Business Need

CivicConnect is intended to improve communication between citizens and the organisations responsible for providing public services. At present, communication and service requests can be fragmented or inefficient, making it difficult for citizens to submit requests, receive feedback and keep track of their progress. This can also make it more difficult for organisations to manage requests consistently and ensure that the correct information reaches the appropriate staff.

CivicConnect aims to address this by providing a centralised digital platform where citizens can submit relevant requests or information, monitor their progress and receive feedback. From the organisation's perspective, the platform will provide a more organised way of receiving, reviewing and managing citizen interactions, while ensuring that authorised staff have access to the information they need.

The value of CivicConnect therefore goes beyond simply introducing a new software system. Its purpose is to create a more effective connection between citizens and service providers, with the aim of improving accessibility, efficiency, transparency and accountability in the delivery and management of public services.

2. Stakeholders Analysis

Stakeholder	Primary Need / Expectation	Influence	Interest
Citizens / End Users	Simple and accessible way to interact with CivicConnect, submit information/requests and receive feedback	Medium	High
Municipality / Organisation	Reliable platform that supports communication and effective service management	High	High
System Administrators	Ability to manage users, permissions and administrative information	High	Medium
Operational Staff	Accurate and organised information that allows requests to be processed and updated efficiently	Medium	High
Project Client / Representative	System must satisfy the agreed business needs, scope, quality expectations and project constraints	High	High
Management / oversight	Identify outstanding and overdue work and accountable request owners	High	High

3. Main Stakeholder Needs

Stakeholder	Basic Need / Expectation
Citizens / end users	An easy-to-use way to interact with CivicConnect and receive feedback or updates.
Staff	Organised, accurate information that allows requests or issues to be managed efficiently.

Stakeholder	Basic Need / Expectation
System administrators	Control over users, system information and administrative functions.
Responsible organisation	A reliable system that can improve communication and service delivery.
Management / oversight	Useful service-activity information for accountability and performance analysis.

4. Basic Scope

In Scope:

In-Scope Capability	Reason
User access to CivicConnect	Enables identified users to interact with the system
Submission of relevant requests/information	Supports the primary interaction between citizens and the organisation
Viewing submission/request status	Provides users with feedback and improves transparency
Staff access to relevant submissions	Allows authorised personnel to process citizen interactions
Updating relevant request/status information	Allows submissions to progress through the required process
Basic user/role administration	Provides appropriate control over system access
Input validation	Supports information quality and reduces invalid submissions
Controlled categories	Consistent request classification
History and lifecycle feedback	Earlier requests and accepted/rejected/completed outcomes
Staff search, details and assignment	Find work and identify accountable ownership
Controlled transitions and closure	Valid updates and authorised completion
Basic oversight	Open, overdue, resolved and closed work by category/status

Out of Scope:

Out-of-Scope Item	Justification
Functionality unrelated to CivicConnect's defined business problem	Prevents unnecessary expansion of project scope
Replacement of unrelated external organisational systems	CivicConnect should address its defined problem rather than replace every existing system
Production deployment during M1	M1 establishes the engineering foundation rather than production implementation
Final architecture and technology selection during M1	These decisions require later evidence and analysis
Detailed database/API/UI implementation during M1	These belong to later lifecycle activities

Deferred / Future Scope:

Deferred Item	Reason for Deferral
Advanced reporting/analytics	Core operational requirements should be established first
Additional external-system integrations	Integration requirements and external dependencies require further evidence
Dedicated mobile application	Need, cost and technical implications should first be evaluated
Advanced automated notifications	Exact stakeholder and communication requirements must first be confirmed
Additional enhancement features	Should only enter scope after stakeholder value and project impact are analysed

5. Possible Stakeholder Conflict

Citizens may want CivicConnect to be very simple and quick to use, while administrators and the responsible organisation may require additional information, validation and security. The project will therefore need to balance ease of use with sufficient control, security and data quality.

6. Functional Requirements (FRs)

The functional requirements below define the core capabilities CivicConnect must provide. Each requirement is uniquely identified, prioritised, linked to its originating stakeholder/source, and paired with measurable acceptance criteria so that it can be objectively verified later in the project lifecycle.

ID	Functional Requirement	Priority	Source / Stakeholder	Acceptance Criteria
FR-01	The system shall allow a citizen to register and log in with a unique account before submitting a request.	Must Have	Citizens / End Users	Valid unique registration creates one account and permits login. Duplicate identifiers are rejected. Invalid credentials create no authenticated session; valid credentials access the correct account.
FR-02	The system shall allow a citizen to submit a service request or piece of information through a structured form.	Must Have	Citizens / End Users (Problem & Business Need)	A valid title, description, location and active category are saved with requester identity and submission time; a unique reference is displayed. Missing or invalid required values create no request. See Section 19.
FR-03	The system shall allow a requester to view a history of all their submitted requests and each current status.	Must Have	Citizens / End Users	Given three own requests in different states, the requester can list and open all three with correct statuses. Another requester cannot access them through the UI or direct requests.
FR-04	The system shall allow authorised operational staff to view submissions assigned to them or their department.	Must Have	Operational Staff	A staff member logged in with the Staff role sees only submissions relevant to their department; unauthorised submissions are not visible or accessible.
FR-05	The system shall allow authorised staff to update the status and add remarks to a citizen's request.	Must Have	Operational Staff	A permitted update saves status, remark, actor and timestamp. Refreshing within 5 seconds shows the saved state to the requester. Invalid transitions do not alter the record; see FR-13.
FR-06	The system shall allow administrators to create, edit and deactivate accounts and assign roles, with updated access enforced on subsequent authenticated requests.	Must Have	System Administrators	An administrator can create an account, edit permitted details, deactivate it and assign/change roles. Deactivation and role changes take effect by the next authenticated request. Non-administrators cannot invoke these operations.
FR-07	The system shall validate mandatory fields and reject a request submission if required information is missing or malformed.	Must Have	Citizens / End Users; Municipality / Organisation	Missing or invalid title, description, location or category values are rejected with field-level errors. The same rules apply to direct submissions. No invalid record is saved; limits are defined in Section 19.
FR-08	The system shall give requesters meaningful feedback when a request is accepted, rejected, updated or completed.	Must Have	Master brief Section 3 requester capabilities	Within 60 seconds of a saved lifecycle event, authenticated request history/details show status, event time and a requester-visible explanation. Rejection and resolution require explanations. Email/push transport remains optional.
FR-09	The system shall provide authorised management users with basic service activity and accountability information.	Must Have	Master brief Section 3 management capabilities	For a known dataset and selected submission-date range, counts and lists correctly identify open, overdue, resolved and closed requests by category/status, with assignee and age visible. Unauthorised access is

ID	Functional Requirement	Priority	Source / Stakeholder	Acceptance Criteria
				denied. See Section 19 definitions.
FR-10	The system shall categorise requests using an active controlled category.	Must Have	Master brief Section 3 requester capabilities	Unknown/inactive categories are rejected. Administrators maintain the list; retirement preserves historical request category labels.
FR-11	The system shall let authorised staff search, filter and sort relevant requests and view full details.	Must Have	Master brief Section 3 staff capabilities	Reference/title search and category/status/assignee filters return matching authorised records. Submitted-date/due-date sorting is correct. Details show description, location, owner, times and lifecycle actions; inaccessible records remain hidden.
FR-12	The system shall let authorised staff assign or accept responsibility for a request.	Must Have	Master brief Section 3 staff capabilities	Staff can accept unassigned work or assign an eligible staff member. Assignee, actor and time are recorded. Ineligible/unauthorised assignments are rejected without changing ownership.
FR-13	The system shall enforce controlled status transitions and record actions and resolution information.	Must Have	Master brief Section 3 staff capabilities	Permit Submitted to Accepted or Rejected; Accepted to In Progress; In Progress to Resolved; Resolved to Closed. Only authorised staff transition records. Acceptance requires an assignee and due time; rejection requires a reason and resolution requires a summary. Record actor/time/explanation. Reject all other transitions.

7. Non-Functional Requirements (NFRs)

Non-functional requirements describe the quality attributes CivicConnect must meet. They are measurable rather than descriptive, and directly inform later architecture and technology decisions without pre-empting them at this stage.

ID	Non-Functional Requirement	Priority	Source / Stakeholder	Acceptance Criteria
NFR-01	Usability: The citizen-facing submission process shall be simple enough for a first-time user to complete without assistance.	Must Have	Citizens / End Users (Possible Stakeholder Conflict)	With five first-time representative users and the same valid scenario, at least four submit successfully within 5 minutes without assistance. Record individual times, errors and feedback.
NFR-02	Security: Access to citizen personal information and request data shall be restricted according to the authenticated user's role.	Must Have	System Administrators; Municipality / Organisation	A user with the Citizen role cannot view, via the UI or a direct request, another citizen's submissions or personal data; role-based access is enforced on every relevant request.
NFR-03	Performance: Status lookup shall meet the defined response-time target under the specified test load.	Should Have	Citizens / End Users; Operational Staff	With 50 concurrent authenticated users and 10,000 requests, each user performs one lookup every 5 seconds for 10 minutes after warm-up. At least 95% complete within 3 seconds; fewer than 1% fail. Record hardware/network conditions.

ID	Non-Functional Requirement	Priority	Source / Stakeholder	Acceptance Criteria
NFR-04	Availability: The platform shall be available for citizen use during standard business hours with minimal unplanned downtime.	Should Have	Municipality / Organisation; Project Client / Representative	Over 20 weekdays, achieve at least 99% availability during 08:00–17:00 Africa/Johannesburg, measured by one-minute probes. Exclude only maintenance announced 24 hours ahead and report excluded minutes separately.
NFR-05	Data Quality: The system shall enforce validation rules on all citizen-submitted data to reduce incomplete or malformed records.	Should Have	Operational Staff; Municipality / Organisation	All valid/invalid boundary cases for submission fields pass validation tests. An audit of 100 saved requests (or all if fewer) finds no missing mandatory or invalid-category values.
NFR-06	Maintainability: The system's requirements, decisions and changes shall be traceable to support controlled maintenance after M1.	Should Have	Project Client / Representative (SEN381 Master Project Brief)	Each baselined requirement can be traced, via the RTM, to its source, its acceptance criteria and any linked design or test evidence added in later milestones.
NFR-07	Scalability: The design shall support defined growth in users and requests while preserving lookup behaviour.	Could Have	Municipality / Organisation	At M2 assess 50,000 stored requests and 100 concurrent users against the same lookup pattern and 3-second 95th-percentile target. Record capacity assumptions and bottlenecks; load-test at M3. Resource/configuration increases are permitted with cost recorded.

Prioritisation note: Requirements are prioritised using MoSCoW (Must Have / Should Have / Could Have). "Must Have" requirements are essential to the core problem defined in Section 1 and the in-scope capabilities defined in Section 4; "Should Have" requirements materially improve quality or usability but would not block a minimal working baseline; "Could Have" requirements add value but are the first candidates for deferral if constraints (schedule, resources) require trade-offs.

8. Initial Requirements Traceability Matrix (RTM)

The RTM below establishes live traceability from stakeholder/source through to each requirement and its acceptance criteria, and provides the structure to which later design, change and test evidence will be linked. It will be maintained and extended as a controlled artefact throughout the project.

Req ID	Source / Stakeholder	Requirement Summary	Acceptance Criteria Ref.	Design Evidence	Test Evidence	Status
FR-01	Citizens / End Users	Account registration and login	FR-01 AC	Pending (M2)	Pending (M2)	Approved for review
FR-02	Citizens / End Users	Submit a service request	FR-02 AC	Pending (M2)	Pending (M2)	Approved for review
FR-03	Citizens / End Users	Own request history and status	FR-03 AC	Pending (M2)	Pending (M2)	Approved for review
FR-04	Operational Staff	View assigned submissions	FR-04 AC	Pending (M2)	Pending (M2)	Approved for review

Req ID	Source / Stakeholder	Requirement Summary	Acceptance Criteria Ref.	Design Evidence	Test Evidence	Status
FR-05	Operational Staff	Update request status/remarks	FR-05 AC	Pending (M2)	Pending (M2)	Approved for review
FR-06	System Administrators	Manage user accounts and roles	FR-06 AC	Pending (M2)	Pending (M2)	Approved for review
FR-07	Citizens / Organisation	Validate mandatory submission fields	FR-07 AC	Pending (M2)	Pending (M2)	Approved for review
FR-08	Citizens / End Users	Notify citizen of status change	FR-08 AC	Pending (M2)	Pending (M2)	Approved for review
FR-09	Organisation / Client	Basic request summary view	FR-09 AC	Pending M2	Pending M3	Approved for review
NFR-01	Citizens / End Users	Usable submission process	NFR-01 AC	Pending (M2)	Pending (M2)	Approved for review
NFR-02	Administrators / Organisation	Role-based access to data	NFR-02 AC	Pending (M2)	Pending (M2)	Approved for review
NFR-03	Citizens / Staff	Response time under normal load	NFR-03 AC	Pending (M2)	Pending (M2)	Approved for review
NFR-04	Organisation / Client	Platform availability	NFR-04 AC	Pending (M2)	Pending (M2)	Approved for review
NFR-05	Staff / Organisation	Data validation quality	NFR-05 AC	Pending (M2)	Pending (M2)	Approved for review
NFR-06	Project Client / Master Brief	Traceable, maintainable baseline	NFR-06 AC	This RTM	N/A at M1	Approved for review
NFR-07	Organisation	Scalable data/access approach	NFR-07 AC	Pending M2	Pending M3	Approved for review
FR-10	Master brief Section 3 requester capabilities	The system shall categorise requests using an active controlled category.	FR-10 AC	Pending M2	Pending M3	Approved for review
FR-11	Master brief Section 3 staff capabilities	The system shall let authorised staff search, filter and sort relevant requests and view full details.	FR-11 AC	Pending M2	Pending M3	Approved for review
FR-12	Master brief Section 3 staff capabilities	The system shall let authorised staff assign or accept responsibility for a request.	FR-12 AC	Pending M2	Pending M3	Approved for review
FR-13	Master brief Section 3	The system shall enforce	FR-13 AC	Pending M2	Pending M3	Approved for review

Req ID	Source / Stakeholder	Requirement Summary	Acceptance Criteria Ref.	Design Evidence	Test Evidence	Status
	staff capabilities	controlled status transitions and record actions and resolution information.				

"Design Evidence" and "Test Evidence" are intentionally recorded as Pending or Deferred candidate at M1, since detailed design and test execution fall outside the M1 boundary (Section 5 of the Milestone brief). Recording the structure now rather than leaving it absent is what allows later architecture, construction and verification evidence to be linked back to a specific requirement without re-engineering traceability retroactively.

9. Traced Requirement Example: Stakeholder → Requirement → Acceptance Criteria

To demonstrate that the RTM is genuinely usable and not only a table of identifiers, one requirement is traced end-to-end below, from its stakeholder source through to its measurable acceptance criteria.

Traceability Step	Detail
Source / Stakeholder	Citizens / End Users — identified in the Stakeholder Analysis (Section 2) as needing "a simple and accessible way to interact with CivicConnect, submit information/requests and receive feedback".
Business Need Link	Directly supports the Problem/Business Need (Section 1): citizens currently struggle to submit requests and track their progress; CivicConnect must let them do both centrally.
Scope Link	Corresponds to the in-scope capability "Submission of relevant requests/information" and "Viewing submission/request status" in the Basic Scope (Section 4).
Requirement	FR-02 — The system shall allow a citizen to submit a service request or piece of information through a structured form.
Acceptance Criteria	A citizen can complete and submit a request containing all mandatory fields; on submission the system displays a confirmation reference number.
Related Requirements	Linked to FR-01 (a citizen must be registered/logged in to submit), FR-03 (the citizen must then be able to view the status of what was submitted) and FR-07 (mandatory fields must be validated before the submission is accepted).
Evidence Still to be Added	Design evidence (submission form design/data model) and test evidence (functional test confirming the confirmation reference is generated and stored) will be linked to FR-02 in the RTM from M2 onward, once architecture and construction begin.

This trace shows the full chain currently available at M1: a named stakeholder need feeds a scoped capability, which is expressed as a uniquely identified, prioritised requirement with testable acceptance criteria, and is positioned in the RTM ready to receive design and test evidence in later milestones.

10. Project Constraints

This section records the formal constraints within which CivicConnect must be engineered and identifies their engineering implications. Constraints are not merely administrative boundaries; they directly influence requirements prioritization, scope decisions, and later architectural choices.

10.1 Scope Constraints

The scope baseline defined in Section 4 of this PED represents the formal constraint on what will be delivered. The explicit out-of-scope and deferred items are controlled constraints designed to prevent scope creep and preserve the project's focus on its core business need.

Engineering Implication: The team must prioritize features aligned with the "Must Have" and "Should Have" requirements identified in Section 6. Any request for work falling outside the formal scope baseline requires a formal change process through the Engineering Decision Log and reassessment of all other constraints. This directly supports the risk mitigation strategy for R-01 (Scope Creep).

10.2 Schedule Constraints

The project follows the SEN381 semester schedule, with Milestone 1 establishing the engineering baseline by the due date. The overall project lifecycle is bounded by the academic calendar, meaning time to complete M1 (Engineering Foundation), M2 (Architecture), and M3 (Construction and Verification), and M4 (Final Product and Engineering Defence) is fixed.

Engineering Implication: This schedule constraint forces the team to make deliberate scope decisions and to avoid premature implementation. The phased milestone approach is the primary control mechanism to ensure completion within the fixed period. This constraint supports deferring advanced analytics and implementation choices, while preserving all mandatory business capabilities including basic reporting.

Interaction with Other Constraints: The fixed schedule interacts with the cost/resource constraint. With a four-person team and a fixed deadline, the project must control scope and avoid over-engineering. This reinforces the need for the MoSCoW prioritization applied in the requirements section.

10.3 Cost and Resource Constraints

The project is delivered by a team of four students, with no external budget for third-party services, infrastructure, or tools beyond free-tier or university-provided resources. The team’s available effort is bounded by other academic commitments.

Engineering Implication: At M2 the team will compare viable technology options against cost, licensing, capability and environment constraints; no final stack is selected in M1. The team will coordinate shared work through GitHub issues and the Team Working Agreement. All team members must understand and be able to defend the project artefacts, as required by the assessment.

10.4 Quality Constraints

CivicConnect must be accepted by the assessor, who will evaluate the project against the criteria in the M1 brief and the SEN381 CivicConnect Master Project Brief. Quality is measured not only by functionality but by the presence of a controlled, traceable, and defensible engineering baseline.

Engineering Implication: The project must produce high-quality controlled artefacts (PED, RTM, Risk Register, Decision Log, AI Usage Register). The team cannot rely on "just getting it to work"; the engineering process and evidence are as important as the final code. This constraint is why NFR-06 (Maintainability) requires traceability and why the RTM has been established as a live artefact.

10.5 Security Constraints

CivicConnect manages potentially sensitive citizen information and requests. The system must therefore enforce role-based access control, ensure data confidentiality, and protect against common vulnerabilities from the outset.

Engineering Implication: Security cannot be an afterthought. The system design (in M2) must explicitly incorporate authentication, authorization, and data validation to satisfy NFR-02 (Access Control). The security constraints directly influence why FR-06 (User Management) is prioritized as a "Must Have" requirement—without it, role-based access control cannot be enforced. This constraint also influences the Forward Engineering Consideration of User Data Privacy (Section 12.6).

Interaction with Scope Constraints: The security constraint forces the system to have robust user and role management. This means that what might seem like an optional administrative feature is actually essential for compliance with NFR-02. The scope must therefore include adequate user management functionality.

10.6 Constraint Trade-offs and Interactions

A key engineering responsibility is to recognize that constraints do not exist in isolation. They interact and managing them often involves trade-offs.

Constraint Pair	Nature of Interaction	Engineering Decision
Quality vs. Schedule & Resources	The project has a strict schedule and a small team. Achieving the highest possible quality on all requirements is not feasible.	Preserve mandatory capabilities and quality targets; defer advanced enhancements when capacity is constrained.
Security vs. Usability	Citizens want a simple, quick process, but administrators require sufficient information, validation, and security.	The team has balanced this by ensuring mandatory fields are minimal yet sufficient and that security controls are enforced transparently in the background.
Scope vs. Resources	The team is small and cannot deliver every feature.	Scope Baseline explicitly excludes functionality unrelated to the core

Constraint Pair	Nature of Interaction	Engineering Decision
		business problem and defers advanced features for future consideration.
Security vs. Schedule	Implementing robust security controls takes time and careful design.	Security requirements are designated "Must Have" to ensure they are addressed early, preventing costly rework later. This aligns with the Forward Engineering Consideration on User Data Privacy.

The principal trade-off preserves minimum business capabilities and their acceptance criteria while deferring advanced analytics, additional integrations and a dedicated mobile application. Priority affects delivery order, not the quality required of delivered functionality. Basic reporting, validation and lifecycle feedback remain mandatory.

11. Initial Risk Register

The Risk Register below identifies project-specific risks that could threaten the successful completion of CivicConnect. Each risk is tested for probability and impact, assigned a priority, and provided with mitigation and contingency plans. The register is a live artefact that will be updated as the project progresses.

ID	Risk Description	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
R-01	Scope Creep: Uncontrolled addition of features or requirements beyond the baseline due to unapproved feature additions or misinterpretation of stakeholder needs.	Medium	High	High	Strict adherence to the Scope Baseline; formal review of all pull requests to ensure alignment with FRs; use of "Deferred/Future Scope" list for new ideas; all changes must be recorded in the Engineering Decision Log.	Re-baseline the project with assessor approval; deprioritize existing "Should Have" features to accommodate new work; escalate to lecturer if necessary.	Project team	Active
R-02	Schedule Overrun: Failure to complete M1 deliverables on time due to inefficient team communication, unclear task allocation, or underestimation of effort.	Medium	High	High	Create and track GitHub issues for all tasks; adhere to Team Working Agreement; identify blockers early in team meetings; maintain regular communication through the team chat channel.	Re-prioritize tasks to ensure core deliverables (PED, RTM, Registers) are completed; temporarily reduce non-essential artefact polish; request a brief extension if justified.	Project team	Active

ID	Risk Description	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
R-03	Technical Complexity: The chosen technology for M2/M3 may prove too complex for the team to implement effectively within schedule constraints, impacting quality.	Medium	Medium	Medium	Conduct a fair technology evaluation in M2; choose a technology with at least some team familiarity; build a small proof-of-concept early before committing fully.	Select an alternative, simpler technology; reduce the scope of features in M3 to match team capability; seek guidance from the lecturer.	Project team	Active
R-04	Poor Traceability: The link between requirements, design, and testing evidence becomes broken or unclear, losing the controlled engineering baseline.	Low	High	Medium	Maintain the RTM as a live artefact from M1 onward; update the RTM with design and test evidence in M2 and M3 as soon as it is produced; cross-reference design and test artefacts in commits; use consistent unique IDs.	Audit the RTM and evidence in the final days of M2/M3; add missing links before submission; designate a team member to oversee traceability.	Project team	Active
R-05	Inadequate Testing: Critical requirements are not adequately tested, leading to a working but unreliable system that fails to meet acceptance criteria.	Low	High	Medium	Write test cases derived directly from acceptance criteria at the same time as design; use the RTM to drive test planning; ensure all "Must Have" requirements have corresponding tests.	Perform a final "smoke test" against the RTM before submission; document any known defects transparently; prioritize testing of "Must Have" features.	Project team	Active
R-06	AI Misuse: AI-generated content could introduce errors, inconsistencies, or academic integrity issues if it is not checked against project evidence.	Low	Medium	Low	The project team checked and verified all recorded AI-assisted work against the relevant source material. Maintain the single AI Usage Register and retain supporting evidence for each entry.	The project team reviewed the AI-assisted work, corrected discrepancies before approval, and retained evidence of verification. The register records each decision and scope boundary.	Project team	Controlled
R-07	Stakeholder Misalignment: The team's interpretation of stakeholder needs differs from the	Medium	High	High	Maintain clear traceability from stakeholder needs to requirements (Section 9);	Revisit stakeholder analysis and adjust requirements; update the RTM and	Project team	Active

ID	Risk Description	Probability	Impact	Priority	Mitigation	Contingency	Owner	Status
	client's actual expectations, resulting in a system that does not meet its intended purpose.				use the RTM to demonstrate the link; seek clarification from the lecturer (acting as client) if ambiguous requirements arise.	associated artefacts; communicate changes through the Engineering Decision Log.		
R-08	Citizen data exposure caused by incomplete access/data-handling rules.	Medium	High	High	Least privilege, data minimisation, synthetic test data and negative-access tests.	Restrict access, preserve evidence, assess exposure and fix before release.	Project team	Active
R-09	Deployment blocked by assumed hosting access, quotas or costs.	Medium	High	High	Validate environment, cost and recovery options in M2.	Use an evaluated alternative; record schedule/scope impact.	Project team	Active

R-01, R-02 and R-07 initially have Medium probability and High impact, giving High priority. R-08 and R-09 are additional High-priority risks. These ordinal ratings prioritise action; they do not establish certainty or catastrophic impact. Owners review ratings at each milestone.

R-01 (Scope Creep) is particularly dangerous because it directly undermines the controlled engineering baseline that M1 is meant to establish. If the team starts adding features beyond what is defined in the Scope Baseline, the requirements will no longer be stable, the RTM will become inaccurate, and the project will lose its controlled foundation. This is why the Scope Baseline includes a deliberate "Deferred/Future Scope" list—to provide a place for new ideas without compromising the current baseline.

R-07 (Stakeholder Misalignment) is equally critical because the entire project is built upon the stakeholder analysis. If the team has misinterpreted what citizens, staff, or the organization actually need, then all subsequent requirements, design decisions, and testing efforts will be misdirected. The RTM traceability example is the primary control against this risk, as it demonstrates a clear chain from stakeholder need to acceptance criteria.

R-02 (Schedule Overrun) is prioritized because failure to complete M1 on time would prevent the team from presenting and defending their work, which is a compulsory requirement for every registered student. The team's use of GitHub issues and protected-main governance are the primary mitigations against this risk.

12. Forward Engineering Considerations

The following seven later-lifecycle concerns require attention during M1, without prematurely making decisions or committing to a final solution. These considerations directly influence the requirements, constraints, and risks already identified in this PED. For each concern, the team has considered why it matters now, what later decisions it may influence, what information is still missing, and the risk of ignoring it.

12.1 Technology Stack Selection

The choice of programming language, web framework, and database will heavily influence the system's performance, security, and maintainability, directly impacting NFR-02 (Access Control), NFR-03 (Response Time), and NFR-07 (Scalability). This decision also affects the team's ability to implement requirements within the cost/resource and schedule constraints. The technology stack will influence the M2 architecture design, M3 construction, deployment environment, and operational costs. At this stage, however, the team lacks a detailed evaluation of available technologies against the NFRs, as well as clarity on deployment environment restrictions and team familiarity with potential options. Ignoring this concern could result in selecting a technology in M2 without proper consideration, leading to technical debt, poor performance, or a system that is difficult to test or deploy. This risk is already captured as R-03 (Technical Complexity).

12.2 Automated Testing Strategy

The testability of the system should be considered alongside the requirements themselves. The NFRs, particularly NFR-03 on response time and NFR-06 on maintainability, need to be testable, which may require specific tools or design patterns. Testing is essential for verifying acceptance criteria and catching defects early. This concern will influence test planning in M3, code structure, and CI pipeline design. The approach to testing may also influence technology selection. Currently, the team has not yet determined the exact testing framework, the balance between unit, integration, and end-to-end testing, or the level of test coverage required to satisfy M3 expectations. Without a clear testing strategy, the team risks building a system that is difficult to test, leading to low-quality code that does not meet its acceptance criteria (R-05). This could also reduce the team's ability to demonstrate traceability from requirements to test evidence.

12.3 Deployment Environment and Operations

The target deployment environment—whether cloud, local, or university server—and the operational requirements (availability NFR-04) impose constraints on the design, architecture, and technology choices. The system must be deployable within the project's resource constraints. This decision will influence technology choice, architecture, scalability (NFR-07), and the CI/CD pipeline. However, the team does not yet know the specific deployment target, operational access rights available, or university policies on hosting. Ignoring this concern could result in a system that cannot be easily deployed to the target environment, or one that lacks necessary capabilities such as logging and monitoring required for operation. This aligns with the schedule constraint, as deployment issues could cause delays in M3.

12.4 Data Migration and Recovery

The system will persist data including requests, user accounts, and status updates. The team must therefore consider how data migration will be handled if the data model changes, and what backup and recovery plan will ensure data integrity and availability. These are quality and security concerns. Data migration and recovery will influence database schema design, testing, and operational procedures. At present, the team does not have clarity on the initial data set, the need for data backup or

migration scripts, or the acceptable recovery time objective. Ignoring this concern could result in data loss during updates, or the system failing to recover from a database failure, which would lose citizen requests. Such a failure would violate the quality and security constraints and directly relate to NFR-04 (Availability).

12.5 Observability and Logging

Once the system is in operation, or even during testing, the team needs to understand what is happening within the system. This includes logging user actions, system errors, and performance metrics to debug issues and ensure security. This is directly relevant to NFR-02 (Access Control) and operational quality. Observability will influence architecture design, logging frameworks, deployment, and operations. The level of logging may also affect performance (NFR-03). The team still needs to determine the level of logging detail required, the tools to be used, and storage and retention policies for logs. Without adequate logging and monitoring, a system that crashes or behaves unexpectedly becomes very difficult to debug, potentially leading to project failure or schedule overrun (R-02). Security breaches may also go undetected.

12.6 User Data Privacy

The system handles citizen data including requests and personal information, which raises legal and ethical privacy concerns. This is directly related to NFR-02 (Access Control), the Security Constraints, and the stakeholder conflict between citizen simplicity and organisational control. Privacy concerns will influence architecture, security design, technology selection, and data handling policies. The team does not yet have clarity on specific university, legal, or ethical requirements for handling user data, or on anonymisation or data minimisation requirements. Ignoring this concern could result in a system that is non-compliant with data protection principles, leading to ethical and potential legal issues, not to mention a failing grade. This also reinforces the high priority of R-07 (Stakeholder Misalignment), as the organisation likely has specific privacy expectations that must be understood.

12.7 Maintainability and Technical Debt

The system is not a one-off project; the team must build a baseline that is maintainable and traceable to support later changes, as captured in NFR-06. Taking shortcuts now will create technical debt that makes future milestones more difficult and costly. Maintainability will influence code quality, documentation, and RTM traceability, as well as the team's ability to respond to scope changes (R-01). The team still needs to establish clear coding standards and review processes and determine the level of documentation required for maintainability. If maintainability is ignored, the codebase could become messy increasing the time for bug fixes, feature additions, and test execution in M3. This risks schedule overrun (R-02) and poor quality (R-05). The protected-main branch and two-reviewer approval governance model serve as the primary early controls against this risk.

13. Relationship to Other Artefacts

This section shows how constraints, risks and forward engineering considerations connect to the overall project work.

Artefact	Relationship
Requirements	The Security Constraint justifies why NFR-02 (Access Control) and FR-06 (User Management) are "Must Have" requirements. The Schedule Constraint defers advanced analytics while preserving FR-09 basic oversight.
Scope Baseline	The Scope Constraint reinforces the in-scope, out-of-scope, and deferred approved scope decisions. The trade-off between Quality and Schedule supports the scope exclusions.
Stakeholder Analysis	The stakeholder conflict identified in Section 5 is reflected in the Usability vs. Security trade-off. R-07 (Stakeholder Misalignment) directly references the stakeholder analysis.
RTM	R-04 (Poor Traceability) is mitigated by the existence of the RTM and the traceability example in Section 9. The Forward Engineering Consideration on Maintainability reinforces the need for the RTM to remain a live artefact.
Engineering Decision Log	Any decision to change scope or accept a risk will be recorded in the Decision Log, as required by the Master Project Brief. The trade-offs identified will also be referenced in the Decision Log.

Technology – Decisions, GitHub & Project Control

Milestone 1 controlled artefacts and governance evidence

This section establishes the project-control evidence required for Milestone 1: the Engineering Decision Log, repository governance model, issue/backlog structure, branch and commit rules, Pull Request and review controls, AI Usage Register, PED v1.0 document control, team review coordination and the baseline gate. The controls below are grounded in the existing CivicConnect PED and the governing SEN381 project briefs.

14. Engineering Decision Log

The Engineering Decision Log records significant project decisions so that the team can later explain the context, constraints, alternatives, rationale, trade-offs, risks, supporting evidence and downstream consequences. Draft decisions require dated team confirmation before being treated as accepted. Choices that require evidence from M2 or later are deliberately marked as deferred rather than being guessed prematurely.

Decision ID	Decision / Deferred Item	Status	Current Evidence
EDL-001	Core M1 scope baseline	Team confirmation pending	Sections 4 and 10.1; R-01
EDL-002	MoSCoW prioritisation and controlled deferral	Team confirmation pending	Sections 6–8 and 10.6
EDL-003	Security/usability balance	Team confirmation pending	Section 5; FR-06/FR-07; NFR-01/NFR-02
EDL-004	Mandatory basic oversight; advanced analytics deferred	Approved correction	Master Section 3; FR-09; review 2026-09-09
EDL-005	Final technology stack and software architecture	Deferred	Sections 4 and 12.1
EDL-006	Detailed database, API and UI design/implementation	Deferred	Sections 4 and 12
EDL-007	Deployment target, CI/testing tooling and notification mechanism	Deferred	Sections 12.2–12.3; FR-08

EDL-001 – Core M1 scope baseline

Decision ID / Status	EDL-001 / Recorded draft decision; Approved
Context	CivicConnect must address the service-request communication problem without expanding M1 into implementation or unrelated organisational systems.
Constraints	Fixed milestone schedule, limited team resources, controlled scope, quality and security obligations.
Alternatives	Broader feature set; replacement of external organisational systems; production implementation during M1.
Decision	Commit M1 to the core service-request lifecycle and associated access, status, role-administration and validation capabilities; exclude unrelated systems and production implementation.
Rationale	A smaller controlled baseline preserves traceability and lets the team demonstrate the core stakeholder value before optional enhancements are added.
Trade-offs	The team accepts less breadth in exchange for stronger control, clearer testing targets and reduced schedule risk.
Risks	Scope creep (R-01) if excluded or deferred work is added without change control.
Evidence	Scope Baseline in Section 4; Scope Constraint in Section 10.1; Risk R-01.
Later consequence	Any post-baseline scope addition must be analysed and authorised through change control before the baseline and RTM are updated.

EDL-002 – MoSCoW prioritisation and controlled deferral

Decision ID / Status	EDL-002 / Recorded draft decision; Approved
Context	Not all requirements can receive equal implementation priority within the fixed schedule and resource limits.
Constraints	Schedule, team capacity, quality expectations and stakeholder value.
Alternatives	Treat every requirement as equally mandatory; prioritise informally without an explicit scheme.
Decision	Use MoSCoW while retaining every master-brief minimum capability, including validation, feedback and basic reporting, as Must Have. Defer additional enhancements or implementation choices.
Rationale	The approach makes trade-offs visible and gives the team a controlled basis for deferring lower-value work rather than rushing all features.
Trade-offs	Some useful functionality may be delivered later, but core quality is protected.
Risks	Stakeholder dissatisfaction if deferred items are assumed to be committed; mitigated by the explicit deferred/future scope list.
Evidence	FR/NFR priority tables, prioritisation note, constraint trade-off table and deferred/future scope.
Later consequence	M2 and M3 planning must preserve Must Have coverage first; any priority change must update the requirements and RTM.

EDL-003 – Security/usability balance

Decision ID / Status	EDL-003 / Recorded draft decision; Approved
Context	Citizens need a simple process while the organisation requires validation, role control and protection of citizen information.
Constraints	NFR-01 usability, NFR-02 security, data-quality needs and the stakeholder conflict in Section 5.
Alternatives	Minimise controls for simplicity; require excessive information and intrusive controls for security.
Decision	Keep citizen-facing mandatory input limited to what is necessary while enforcing validation and role-based access controls as part of the system behaviour.
Rationale	The decision protects both accessibility and organisational control rather than treating them as mutually exclusive.
Trade-offs	Additional validation and access checks add design/test effort, but reduce privacy and data-quality risk.
Risks	Poor usability if controls become too intrusive; security/privacy exposure if controls are weakened.
Evidence	Section 5 stakeholder conflict; FR-06, FR-07; NFR-01, NFR-02; Section 10.6 trade-off analysis.
Later consequence	M2 architecture and UI decisions must show how security is enforced without adding unnecessary steps to the citizen workflow.

EDL-004 Basic oversight scope correction

Decision ID / Status	EDL-004 / Approved
Context	The earlier draft treated basic reporting as optional; the 9 September review found mandatory management capabilities in Master Section 3.
Constraints	Minimum capability coverage, schedule and capacity.
Alternatives	Retain basic oversight and defer advanced analytics; alternatively request a formally authorised scope variation. Unapproved omission is not acceptable.
Decision	Propose FR-09 as Must Have; retain advanced analytics as deferred.
Rationale	Aligns scope, requirements, RTM and constraint reasoning with the governing brief.
Trade-offs	Basic reporting requires delivery effort; defer advanced enhancements to protect capacity.
Risks	R-07 if minimum scope is misunderstood; R-02 if reporting effort is omitted.
Evidence	Master brief Section 3; FR-09; Section 4; review dated 2026-09-09.
Later consequence	Team confirms correction before baseline; M2 design and M3 tests cover FR-09. Version history preserves the earlier draft.

EDL-005 – Final technology stack and architecture deliberately deferred

Decision ID / Status	**EDL-005 / Deferred**
Context	The project needs a technology stack and architecture later, but M1 is explicitly a foundation milestone.
Constraints	Technology must be justified against requirements, security, cost, schedule, team capability, deployment and maintainability.
Alternatives	Select a preferred stack immediately; copy a familiar stack without analysis; defer pending evaluation.
Decision	Do not select the final stack or architecture in M1. Carry the decision into M2 after comparing viable options against the baselined requirements and constraints.
Rationale	The team does not yet have sufficient evidence on deployment restrictions, team familiarity, technical trade-offs and architecture consequences.
Trade-offs	Deferral delays implementation certainty but prevents premature lock-in and avoidable technical debt.
Risks	R-03 Technical Complexity if the later selection is not evidence-based.
Evidence	Out-of-scope statement in Section 4; Forward Engineering Consideration 12.1; R-03.
Later consequence	M2 must produce a structured decision/ADR with alternatives, evidence and downstream consequences.

EDL-006 – Detailed database/API/UI decisions deliberately deferred

Decision ID / Status	EDL-006 / Deferred
Context	Persistence, API and UI details will be needed, but they depend on the chosen architecture and refined design evidence.
Constraints	M1 boundary, maintainability, data privacy, testability, security and future deployment choices.
Alternatives	Design/implement these areas now; create exploratory notes only; defer formal decisions.
Decision	Keep detailed database schema, API contracts and UI implementation outside the M1 baseline; record only the requirements, risks and decision criteria that those later designs must satisfy.
Rationale	This preserves design options and avoids creating artefacts that may need to be discarded after M2 architecture decisions.
Trade-offs	Less implementation detail is available now, but the baseline remains lifecycle-appropriate and defensible.
Risks	Later rework if M2 ignores the current NFRs, privacy concerns or RTM links.
Evidence	Out-of-scope statement in Section 4; Forward Engineering Considerations 12.2–12.7.
Later consequence	M2 design artefacts must link back to requirements and update the RTM/Decision Log as decisions are made.

EDL-007 – Deployment, CI/testing tooling and notification mechanism deferred

Decision ID / Status	EDL-007 / Deferred
Context	Deployment, automated verification and citizen notification all influence later engineering, but exact mechanisms are not yet supported by sufficient M1 evidence.
Constraints	Unknown deployment target/access rights, future technology choice, operational cost, testability and stakeholder communication expectations.
Alternatives	Choose tools/channels immediately; keep all options open without recording criteria; defer with explicit evidence needs.
Decision	Defer the exact deployment platform, CI/test toolchain and FR-08 notification channel. Preserve the required outcomes and evaluation criteria in the baseline.
Rationale	The required behaviour is known, but the implementation mechanisms depend on M2 architecture/technology and stakeholder/platform evidence.
Trade-offs	The team cannot yet demonstrate tool-specific implementation, but avoids committing to incompatible or costly options.
Risks	Deployment delay, inadequate testing or communication mismatch if these choices are left too late.

Decision ID / Status	EDL-007 / Deferred
Evidence	Forward Engineering Considerations 12.2 and 12.3; FR-08; deferred advanced notifications.
Later consequence	M2/M3 must convert these deferrals into controlled decisions and link the resulting issue/PR/test/deployment evidence to the RTM.

15. GitHub Repository and Team Governance

GitHub is treated as a configuration-management and engineering-evidence environment. Substantive controlled changes must progress through issues/tasks, branches, meaningful commits and Pull Requests before entering the controlled main branch. The live repository must demonstrate authentic progression rather than reconstructed activity.

Control	Mandatory Expectation	Applied Project Rule / Current Status	Evidence to Retain
Repository	One controlled team repository.	Configured: https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup	https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup
Main branch	Protected and treated as the controlled product state.	Verified 2026-09-09: PR required; 2 approvals; stale approvals dismissed; conversations resolved; administrators cannot bypass; force pushes/deletion disabled.	https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup/settings/branch_protection_rules/82957015
Direct development on main	Not permitted for substantive controlled changes.	Prohibited by team workflow.	Branch protection + history.
Pull Requests	Required for substantive changes entering main.	M1 PR #7 was closed without merge on 2026-09-24; the signed-off v0.11 baseline is recorded separately.	https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup/pull/7

Control	Mandatory Expectation	Applied Project Rule / Current Status	Evidence to Retain
Approval	Minimum TWO approvals from team members other than the author.	Two approvals are required for substantive changes. PR #7 closed without merge on 2026-09-24; no approval or merge is inferred from that PR.	https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup/pull/7
Self-approval	Not accepted.	Prohibited.	Reviewer identities.
Review quality	Meaningful engineering review; no rubber-stamping.	Review checklist defined.	Comments, author response and re-review.
Issues/tasks	Meaningful work represented in backlog/project board and linked where practical.	Existing issues #1–#6 retained. Section 15.2 contains internal task IDs, not live GitHub issue numbers.	https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup/issues
Secrets	No passwords, API keys, tokens, private keys or confidential credentials.	Prohibited; use environment/configuration controls later.	Repository history/settings.
History	Commits, PRs, reviews and merges must be progressive and authentic.	Commits a10c26a and c708aae record real upload/control work. No historical progression or reviews are reconstructed.	Commit/PR timeline.

15.1 Approved Repository Structure

```
/
├── README.md
├── docs/
│   ├── PED/
│   ├── requirements/
│   ├── decisions/
│   ├── risk/
│   ├── change/
│   ├── quality/
│   ├── security/
│   ├── deployment/
│   └── governance/
├── src/
├── tests/
```

```
├── .github/
│   ├── workflows/
│   └── pull_request_template.md
└── .gitignore
```

The structure is intentionally simple. It separates the evolving PED and controlled engineering records from implementation artefacts while preserving links between requirements, decisions, risks, issues/PRs, tests and deployment evidence.

15.2 Internal M1 Task Breakdown

Internal task ID	Task	Team accountability	Definition of Done
M1-01	Baseline problem, stakeholders and scope	Project team	Problem, stakeholder and scope sections reviewed and linked.
M1-02	Functional and non-functional requirements	Project team	Priorities, sources and measurable acceptance criteria complete.
M1-03	Initial RTM and traced requirement example	Project team	All FR/NFR IDs present; one requirement traced end-to-end for M1 evidence.
M1-04	Constraints and trade-off analysis	Project team	Scope, schedule, cost/resources, quality and security implications recorded.
M1-05	Initial Risk Register	Project team	Probability, impact, priority, mitigation, contingency, owner and status complete.
M1-06	Forward Engineering Considerations	Project team	5–7 project-specific concerns with missing evidence and downstream risks.
M1-07	Engineering Decision Log	Project team	Accepted and deferred M1 decisions recorded with evidence and consequences.
M1-08	GitHub governance and protected main	Project team	Main protected; PR-only substantive changes; two non-author approvals.
M1-09	AI Usage Register	Project team	Material AI use recorded and human verification evidence retained.
M1-10	PED v1.0 control, team review and baseline gate	Project team	Version history complete; team review resolved; sign-off recorded.
M1-11	M1 baseline correction pass	Project team	Cross-artefact IDs, scope, RTM, risks, decisions and governance evidence consistent.

15.3 Branch and Commit Standard

- Create a branch for every substantive controlled change. Use descriptive names such as `docs/project-control` or `docs/M1-<task>-<short-name>` for documentation, `feature/FR-XX-<short-name>` for features and `fix/<issue>-<short-name>` for fixes.
- Keep branches focused on one coherent issue or change so that the PR can be reviewed meaningfully.
- Commits must be small enough to explain, use meaningful messages, and reference the relevant issue/requirement/decision where practical.
- Do not use vague messages such as `update`, `final`, `stuff`, `changes` or bulk end-of-milestone commits.

Example commit messages:

```
docs(M1): add initial engineering decision log [M1-07]
docs(RTM): correct FR-08 traceability link [M1-03]
docs(risk): update R-02 mitigation after schedule review [M1-05]
governance: add PR review template and protected-main checklist [M1-08]
```

15.4 Pull Request and Meaningful Review Standard

Every substantive PR must state the purpose of the change, identify linked issues/requirements/risks/decisions where practical, and show what was checked before merge. Reviewers should explicitly consider:

- Alignment with requirements and acceptance criteria.
- Correctness, internal consistency and design/document consistency.
- Maintainability and technical-debt impact.
- Security and privacy implications.
- Test and regression implications, where applicable.
- Dependency changes, where applicable.
- Documentation and traceability impact.
- Whether the change is suitable to enter the controlled baseline.

Required review path:

Review comment → Author response → Correction → Re-review → TWO approvals → Merge

15.5 Protected Main Branch Configuration Checklist

Setting	Configuration	Engineering Purpose
Require a Pull Request before merging	ON	Prevents substantive direct-to-main changes.
Required approving reviews	2	Both approvals must be from team members other than the author.

Setting	Configuration	Engineering Purpose
Dismiss/refresh approval after material changes	Recommended ON	Ensures approval still applies to the current change.
Require conversation resolution before merge	Recommended ON	Prevents unresolved review concerns being bypassed.
Restrict force pushes	ON / not allowed	Protects the controlled history.
Restrict branch deletion	ON / not allowed	Prevents accidental loss of the controlled branch.
Direct commits to main	Not allowed for substantive work	All controlled changes enter through PRs.

Evidence should be presented from the live repository/settings where practical. A screenshot may support the PED, but the live settings, PR history, reviews and issue links are stronger assessment evidence.

16. AI Usage Register

All project AI use is recorded once in this uniform register. Every record uses the same fields for activity, team verification, decision, evidence and scope boundary. AI output supports the work; the project team retains responsibility for its accuracy and approval.

Record (ID, date and tool)	Project activity and AI assistance	Team verification and decision	Evidence and scope boundary
AI-001 Date: 2026-09-09 Tool: ChatGPT	Activity: M1 Engineering Decision Log, GitHub governance, AI register structure, PED document control and baseline gate. AI assistance: Drafted governance structures and decision records, prepared workflows, checklists and templates, and identified unresolved controls.	The project team checked the content against the governing briefs, M1 PED and repository controls. The M1 AI-001 checklist and M1 sign-off are approved. The initial draft gaps in live GitHub evidence and team-size wording were corrected before sign-off. Decision: Approved for the signed-off M1 baseline.	M1 PED v0.11, project briefs, repository rules, PR and review records. AI output was treated as draft material and was checked against project evidence.
AI-002 Date: 2026-09-09 Tool: OpenAI Codex	Activity: Repository setup and project controls. AI assistance: Supported repository review and preparation of the README, PR template, branch and issue controls, PR #7 and related governance records.	The project team checked the changes against source files, Git history, repository rules and merge-block settings. The M1 approval record confirms team review. Decision: Approved; repository history preserves the actual changes.	Source hashes, Git checks, branch-protection settings, PR #7 and repository history. Automated checks are distinguished from the recorded team approval.
AI-003 Date: 2026-09-09 Tool: OpenAI Codex	Activity: M1 content audit and v0.10 corrections. AI assistance: Compared the project briefs, identified missing requirements and targets, and supported corrections to scope, RTM, decisions, team records and evidence references.	The project team checked requirement IDs, scope, structure, links and assumptions against the briefs and controlled PED. The approved M1 v0.11 baseline records the completed review. Decision: Approved after team verification.	M1 brief comparison, approved M1 PED v0.11, RTM, repository-control evidence and review record. Programmatic checks support but do not replace the team review.
AI-M2-001 Date: M2 milestone; approved 2026-09-30 Tool: OpenAI Codex	Activity: M1/M2 comparison, integration records, RTM generation, traceability implementation and automated tests.	The project team checked and verified the generated records, implementation and test results against the approved requirements. The M2 document approval	RTM records, traceability implementation, evidence/verification.txt and retained test output. Automated structural checks and product evidence

Record (ID, date and tool)	Project activity and AI assistance	Team verification and decision	Evidence and scope boundary
	AI assistance: Supported preparation and implementation of the listed project artefacts.	is recorded. Decision: Approved for the M2 project baseline.	are recorded separately.
AI-M2-002 Date: M2 milestone; approved 2026-09-30 Tool: OpenAI Codex	Activity: M2 documentation and evidence audit. AI assistance: Supported reconciliation of the PED, traceability validator, automated checks and repository evidence.	The project team checked and verified the updated document, validator output and evidence. M2 approval is recorded; independent GitHub review records remain part of the merge history. Decision: Approved.	evidence/verification.txt, change records, repository history and M2 review record. Automated results are retained with the project evidence.
AI-M2-003 Date: M2 milestone; approved 2026-09-30 Tool: OpenAI Codex	Activity: Initial lifecycle reference slice and integration update. AI assistance: Supported implementation of the reference slice and its tests.	The project team checked and verified the implementation, test output, demo and integration records against the approved M2 scope. Independent pull-request reviews are recorded separately. Decision: Approved.	Lifecycle service, schema, demo, Product_Trace.md, product_trace.json and retained test output. This evidence applies to the initial reference slice, not the full integrated M3 web application.
AI-M4-001 Date: 2026-10-15; team review 2026-10-16 Tool: Claude (Anthropic)	Activity: M3 staging, performance, registers and release evidence. AI assistance: Supported staging scripts and reports, registers, release summary and PED Sections 39–44. Execution environment: Python 3.12.3, SQLite 3.45.1, Linux sandbox, one CPU core, 4 GB RAM, no network.	The project team checked and verified the integrated M3 requirements, tests, checks and evidence and reran/reviewed the separate lifecycle-slice checks. The team approved the local academic release decision on 2026-10-16. Decision: Approved for local school-project demonstration.	evidence/staging_verification.json, evidence/perf_nfr03_profile.json, load and robustness reports, CI run 38870644636 and main commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659. Slice measurements do not represent integrated website performance; M4 production readiness is no-go.
AI-M4-002 Date: 2026-10-17 Tool: OpenAI Codex	Activity: PED v4.0 closure, final submission preparation and audit. / AI assistance: Supported M1 scope reconciliation, final RTM and calculation, evidence summary, release decision, register updates, final PED content and consistency audit, presentation and speaker guide.	The project team checked and verified the document content, calculations, release decision, evidence references and audit corrections against retained project files and repository records. Supporting evidence is retained in the final package and controlled project repository. Decision: Approved collectively by the project team for final submission; 17 October 2026. Protected-branch approval and merge remain separate governance steps.	Final PED, RTM, scope worksheet, release decision, current register snapshots, retained evidence records and presentation. No application code change or new application test result is claimed for this documentation activity.

17. PED v1.0 Version and Document Control

The Project Engineering Document is a single evolving controlled record. M1 establishes the v1.0 engineering baseline; later milestones update the same PED rather than replacing it with unrelated reports.

Document Title	CivicConnect Project Engineering Document
Document Identifier	PED-CIVIC continuing PED-CIVIC-M1
Planned Baseline Version	v2.0 Architecture Technology and Initial Design Baseline
Current Control State	Approved v2.0.2 with executed initial lifecycle trace.
Document Owner	CivicConnect project team
Governing Documents	M2 brief; project Master Brief and M1 brief must be checked by the team
Controlled Repository Location	Controlled M2 PED and approved pull-request record; version and review history are retained in Governance.md.
Review Frequency	Every material change and before M2 sign-off
Post-baseline Change Rule	Do not silently overwrite baselined content. Approved changes must be linked to change records and affected artefacts/RTM.

17.1 Version History

Version	Date	Purpose / Change Summary	Status
0.9	2026-09-09	Integrated M1 working draft containing existing PED sections plus Technology control artefacts.	Superseded working draft
0.10	2026-09-09	Correct mandatory capabilities, criteria, RTM, team records and GitHub evidence; add AI entries and review agreement.	Superseded review candidate
0.11	2026-09-09	Record the four-person project roster and the shared team responsibility.	Signed-off M1 PED artifact; source for the v1.0 baseline
1.0	2026-09-09	M1 Engineering Foundation and Requirements Baseline; signed-off source artifact is PED v0.11.	Approved baseline
2.0	2026-09-24	Continue M1; add baseline review, change control, RTM, risks, traceability tooling and human review register.	Approved
2.0.1	2026-09-24	Reconcile supplied approval; exact CR matching and	Approved

Version	Date	Purpose / Change Summary	Status
		evidence hashes; ten-task completion audit. No requirement changes.	
2.0.2	2026-09-24	Add executed lifecycle reference slice, 20 product tests, complete trace and consistent RTM links.	Approved

17.2 Controlled Document Rules

- Use stable identifiers for requirements, risks, decisions, changes, tests and defects.
- Preserve version history, authorship, review and approval information.
- Do not silently overwrite baselined scope or requirements after sign-off.
- Link approved changes to the RTM and all affected engineering artefacts.
- Use repository/PR history as the primary evidence of controlled document evolution.
- Support major quality, security, performance and readiness claims with measurable or traceable evidence.

18. Team Review, Baseline Sign-Off and Gate Evidence

Before PED v1.0 is signed off, the registered team should perform a coordinated review across the complete baseline rather than reviewing sections in isolation. The purpose is to verify consistency between stakeholder needs, scope, requirements, RTM, constraints, risks, decisions, repository governance and AI accountability.

18.1 Pre-Sign-Off Review Checklist

Review Area	Review Expectation	Current Readiness
Problem / stakeholder / scope	Present and linked to the CivicConnect business need.	Prepared; Approved
FRs / NFRs / acceptance criteria	Unique IDs, priorities, sources and measurable criteria are present.	Prepared; Approved
RTM	All current requirements represented; traced example available.	Prepared; Approved
Constraints and trade-offs	Scope, schedule, cost/resources, quality and security analysed.	Prepared; Approved

Review Area	Review Expectation	Current Readiness
Risk Register	Probability, impact, priority, mitigation, contingency, owner and status recorded.	Prepared; Approved
Engineering Decision Log	Accepted decisions and justified deferrals recorded with evidence.	Prepared; Approved
GitHub governance	Protected main, PR-only substantive changes, two non-author approvals and progressive evidence.	Settings verified. M1 PR #7 closed without merge; signed-off PED v0.11 is recorded separately.
AI Usage Register	Material AI use recorded with human verification and decision.	AI-001–AI-003 recorded; Approved
PED version/document control	Version history and post-baseline change rules defined.	Prepared; Approved
Team registration / role naming	PED ownership labels match the registered team.	Approved

18.2 Formal Baseline Sign-Off Record

Project	CivicConnect
Baseline Type	Milestone 1 – Engineering Foundation & Requirements Baseline
Version	v1.0
Date	09/09/2026
Scope reviewed	YES
Requirements / traceability checked	YES
Risk review completed	YES
Repository / governance controls checked	YES
Outcome	ACCEPTED

Registered team review / approval evidence:

Review area	Team reviewers	Date	Review Outcome
Traceability	Project team	09/09/2026	Approved
Architecture	Project team	09/09/2026	Approved
Data	Project team	09/09/2026	Approved

Review area	Team reviewers	Date	Review Outcome
Technology	Project team	09/09/2026	Approved

19 Requirement Definitions and Validation Assumptions

The engineering targets and definitions are approved. Any material change follows the controlled-change process.

Topic	Approved definition
Input	Title 5–120 characters; description 10–2000; location 3–200; active category; system-derived identity/time. Trim whitespace; reject blank-only values. M2 defines accessible guidance and safe text handling.
Lifecycle	Submitted → Accepted → In Progress → Resolved → Closed; Submitted → Rejected. Closed/Rejected are terminal. Reopening requires stakeholder clarification and controlled change.
Oversight	Open = Submitted, Accepted or In Progress. Overdue = open with due time passed. Authorised staff set a due time later than submission at acceptance; due-time changes record actor/time/reason. Unaccepted requests without a due time remain visible as untriaged.
Due-time policy	Operational planning dates do not imply contractual service-level agreements. Validate authority and policy with client before baseline.
Categories	Review an initial list such as facilities, maintenance, IT support and security concerns from the scenario; approve the vocabulary before baseline.
Quality targets	Validate NFR-01/03/04/07 targets for stakeholder relevance, feasibility and cost; retain actual decisions and revise linked records if changed.
Sources	Master Section 3 supplies mandatory capabilities; stakeholder needs are interpreted from the scenario. Numerical targets and lifecycle rules are team proposals, not interview evidence.

20 Traceability Extension and Change Control

This companion extends the single RTM in Section 8. Pending means evidence has not yet been produced; no implementation or acceptance is claimed.

Requirement	Change and issue or PR	Implementation	Acceptance and release
FR-01	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-02	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release

Requirement	Change and issue or PR	Implementation	Acceptance and release
FR-03	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-04	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-05	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-06	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-07	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-08	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-09	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-01	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-02	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-03	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-04	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-05	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-06	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
NFR-07	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-10	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-11	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-12	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release
FR-13	v0.10 correction: PR #7; later change IDs pending	Pending M3	Pending acceptance and M4 release

After baseline, record change ID, requester/date, reason/value, affected requirements and design/UI/API/data, security/privacy, tests, scope, schedule/resources, cost and risks; propose accept/modify/defer/reject and record authorised outcome/rationale. Implement on a branch, update RTM and related artefacts, obtain two non-author approvals and identify the new revision. An EDL entry does not replace change-impact analysis.

21 Team Working Agreement and Review Plan

The team working agreement is approved. The team is collectively responsible for explaining and defending the PED.

Team	Shared project responsibility	Identity record
1	The project team shares responsibility for the complete project and all artefacts.	Marco Brazao de Sousa; student number: 601587; GitHub account: to be recorded
2	The project team shares responsibility for the complete project and all artefacts.	Gift Mathe; student number: 600731; GitHub account: to be recorded
3	The project team shares responsibility for the complete project and all artefacts.	Masana Sakala; student number: 600688; GitHub account: to be recorded
4	The project team shares responsibility for the complete project and all artefacts.	Jacobus Petrus Geldenhuys; student number: 602755; GitHub VoidMeBro

Review	Team review approach	Evidence
Scope/requirements	The project team reviews collectively; independent non-author approval is retained where required.	Master Section 3 coverage and acceptance/modification of Section 19
Constraints/risks	The project team reviews collectively; independent non-author approval is retained where required.	Ratings, feasibility, mitigation and accountable owner confirmed
Decision/AI	The project team reviews collectively; independent non-author approval is retained where required.	EDL confirmation and team verification of AI-001 through AI-003
Whole PED/PR	The project team reviews collectively; independent non-author approval is retained where required.	PR links, resolved findings, exact approved SHA and sign-off
Baseline gate	The project team reviews collectively; independent non-author approval is retained where required.	Date/version, outcomes, conditions and genuine evidence

22 Schedule Resource and Risk Controls

Milestone	Planning output	Evidence to enter
M1	Foundation baseline	09/09/2026

Milestone	Planning output	Evidence to enter
M2	Architecture/design and deferred decisions	30/09/2026
M3	Construction/testing and release readiness	Official due date
M4	Product evaluation and defence	Official due date

Risk scale: probability Low = unlikely, Medium = plausible, High = likely without intervention. Impact Low = local rework, Medium = material task/quality delay, High = milestone or mandatory-outcome threat. Priority: Low/Low and Low/Medium = Low; Low/High, Medium/Low, Medium/Medium and High/Low = Medium; Medium/High, High/Medium and High/High = High. Re-rate when evidence changes.

Data revisits EDL-005/006 at M2 design review. EDL-007 platform/tool evidence is gathered in M2 before M3, with Architecture on testing and on feedback outcomes. Escalate missing evidence threatening the next milestone; record date, evidence and outcome when each deferral is resolved.

23 AI Usage Register Control

Section 16 is the sole project AI register. M1–M4 records use the same fields in that register, synchronized with registers/AI_Usage_Register.md. The project team checked and verified each recorded AI-assisted activity against retained project evidence.

24 Baseline Evidence Completion Record

The following checklist preserves the M1/M2 checkpoint state. Later approval and completion evidence is recorded in Sections 33–38, and the current M4 submission gate is recorded in Sections 45–51.

Record	Status and completion field
M1 baseline source	Signed-off M1 baseline: CivicConnect_Milestone1_v0.11.docx; sign-off is recorded separately from PR #7, which closed without merge.
Registration	The four-member team allocation is approved.
Human verification	AI-001–AI-003 approved; review record retained.
Agreement/targets	Four members confirm Section 21; validate Section 19 with client/lecturer.
Approvals	PR #7 closed without merge. M1 and M2 approval evidence is recorded separately in Sections 33–38.

Record	Status and completion field
Visual review	Open corrected DOCX in Word and check every page/table before submission; automated rendering unavailable.
Team sign-off	Complete Section 18.2 for four members with date, exact version and conditions.
Assessor gate	09/09/2026 ACCEPTED

25 References

SEN381 CivicConnect Master Project Brief 2026: Sections 3, 7.1, 8–14, 18 and Appendix D. SEN381 Milestone 1: Sections 3–5 and 9. Supplied CivicConnect M1 draft v0.9: source draft. No unrecorded stakeholder interviews or approvals are claimed.

Repository: https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup

Review: https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup/pull/7

Verified protection: https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup/settings/branch_protection_rules/82957015

Control record: Repository_Control.md.

26 M2 evolution and document control

This team PED integrates document evolution, requirements traceability, change control and the initial lifecycle demonstration. The approved decisions are linked to implementation and verification artefacts. Material additions and changes follow the controlled-change process.

Source basis: supplied M2 brief Sections 4 to 10; supplied M1 v0.11 Sections 6 to 12 and 18 to 24; Assignment 2 Task 1A lifecycle coordination, Task 1B feedback delivery and persistence/integration research; Assignment 1 cover identity record. The A2 recommendations support the approved M2 decisions in Section 37. This pack does not assert independent verification of the papers cited by A2.

Control	Current record
Baseline identity	PED v2.0 is approved in the returned version history and confirmed; v2.0.2 records the completed initial trace and verification.
Preservation	M1_original_v0.11.docx is byte-for-byte source; source_manifest.json records SHA-256. baseline_m1.json preserves all 20 requirement wordings, priorities, sources and criteria.

Control	Current record
Included now	M1 review, changes, risks, 20-row RTM, tooling, executable lifecycle path, 35 tests and presentation evidence.
Project artefacts	The initial product trace is complete. Further application artefacts follow the same requirements and change-control process.
Approval gate	M2 documentation: Approved. Sign-off date: 30/09/2026. The project team confirmed approval. Independent GitHub reviews follow the repository merge process.
Team identity record	The project roster records all four registered team identities; the controlled M1 history is preserved.

27 M1 baseline review and findings

Area	Finding and disposition
Requirements and acceptance	13 FRs and 7 NFRs retained verbatim in the RTM. No numerical target or lifecycle rule is silently changed. All FRs remain Must Have; NFR priorities remain as supplied.
Scope	Basic reporting FR-09, categorisation FR-10, search FR-11, assignment FR-12 and lifecycle FR-13 remain included. Email/SMS, advanced analytics and dedicated mobile work remain deferred.
Baseline status	All M1 v0.11 items are approved. The four-member team sign-off is confirmed. The approved v0.11 baseline is preserved; CR-M2-01 records the document-control update.
Team constraint	The four-member team allocation is confirmed. CR-M2-02 records the team-size clarification for M2.
Acceptance assumptions	Section 19 and all M1 items are approved. The accepted definitions and targets are preserved; subsequent changes follow CR-M2-03 and formal change control.
Engineering status	M1 implementation Pending M3 is historical. M2 now includes the initial lifecycle service and traceability tooling, with 35 passing tests. The RTM records the evidenced lifecycle requirements as In Development; unrelated features remain Not Yet Implemented.
Human notes	M1 AI-002 and AI-003 retain their recorded verification date. M1 v0.11 sign-off is confirmed. GitHub review records and test evidence remain with the project artefacts.
Cross references	M1 uses inconsistent FR-06/NFR-06 spellings in prose; canonical IDs in the RTM are FR-06 and NFR-06. Heading 1.5 Security Constraints appears within Section 10; interpret as historical 10.5, without changing requirement meaning.

28 Controlled change register

Workflow: propose a change and retain the old wording; assess source, acceptance, architecture, data, interfaces, technology, schedule and risks; obtain the required human decisions; implement on a branch; obtain two independent current-revision approvals; update PED/RTM/ADR together; preserve the prior revision. Rejected proposals remain recorded. A Changed RTM status requires an approved change reference; a proposal alone never replaces the original requirement.

ID and subject	Impact and approved action	Decision owner and state
CR-M2-01 M1 review candidate to controlled M2 continuation	Preserve approved v0.11 and the confirmed team sign-off. The M2 continuation and its recorded decisions are approved.	Team; M1 sign-off confirmed; M2 document approved
CR-M2-02 Three versus four team members	Use the confirmed four-person team roster and retain the independent review control.	Approved; clarification recorded 2026-09-24
CR-M2-03 Acceptance and scenario assumptions	M1 values are approved. Any future change must retain old wording and assess criterion, scope, schedule and engineering impacts.	Project team; no requirement changes made; future changes controlled
CR-M2-04 M1 deferred engineering decisions mature for M2	The project team evaluates the deferred architecture, data and technology decisions through the approved ADR process; retain the EDL-007 external-feedback deferral. Evaluate costs, compatibility and data integrity before approval.	Project team; see consolidated project handover

29 Assumptions dependencies and forward engineering

Reference	Current direction and required evidence
A-M2-01 team	The four registered students form the confirmed CivicConnect project team. Planning and delivery are collectively owned.
A-M2-02 resources	The project team documents actual hosting, storage limits and cost evidence before technology approval. R-09/R-15.
A-M2-03 scope	In-app feedback remains sufficient under FR-08; no external notification provider is committed. The project team confirms the history-read contract. EDL-007.
A-M2-04 review	All M1 v0.11 items and the document are approved. The review register records the confirmed four-member sign-off. R-06/R-10.
FEC 12.1 technology	The project team evaluates the wider application stack and deployment selection; EDL-005 remains open for that scope.
FEC 12.2 testing	Verification comprises 15 RTM checks and 20 lifecycle tests, including rollback, concurrent updates and negative-access cases. Further team verification extends this evidence against unchanged acceptance criteria.

Reference	Current direction and required evidence
FEC 12.3 deployment	The project team supplies configuration, secrets boundary, persistent storage, networking and recovery assumptions. Provider and production deployment remain deferred.
FEC 12.4 recovery	The project team defines transactions, migration and restore experiments and approves recovery objectives. No recovery success is claimed.
FEC 12.5 logging	The system separates business history from diagnostics; the project team defines redaction, retention and access. Do not log credentials or citizen payloads by default.
FEC 12.6 privacy	The architecture and data design specify ownership checks, least privilege and minimised test data. Use synthetic citizens for tests; determine applicable data-handling obligations with the project authority.
FEC 12.7 maintainability and technical debt	NFR-06 traceability tooling checks coverage and evidence links. The project team maintains shared coding standards, product-code review and documentation; keep future debt visible in ADR consequences.

30 M2 risk register update

The original R-01 to R-09 remain in Section 11. R-04 covers traceability coordination across the M2 baseline. R-02 now covers M2 evidence integration; R-03 requires stack feasibility evidence; R-08 requires access-control tests; R-09 remains open until deployment compatibility is demonstrated. The documented ordinal assessments and ownership are confirmed; mitigations remain active until evidence closes the risks.

Risk and cause	Rating owner and status	Mitigation and contingency
R-10 M2 approval could be confused with confirmed M1 sign-off	High/High; High priority; Project team; Active	Preserve the confirmed M1 sign-off and approved M2 document identity separately. Independent PR reviewers approve the exact submitted revision before merging.
R-11 Architecture or design differs from actual code	Medium/High; High priority; Project team; Active	Review ADR-to-component mappings in every PR; Reopen ADR and update RTM before baseline
R-12 Partial lifecycle save produces inconsistent feedback	Medium/High; High priority; Project team; Active	Atomic status/history transaction and rollback/concurrency tests; Block affected path until integrity proved
R-13 Framework or dependency cannot run in target environment	Medium/High; High priority; Project team; Active	Version manifest and clean bootstrap on target-like host; Evaluate simpler compatible stack via ADR
R-14 Risk of AI evidence or checklists being accepted without full team review	Medium/High; High priority; Project team; Active	Team verification record, method, findings and evidence per decision; reopen unsupported claims and correct the controlled baseline.

Risk and cause	Rating owner and status	Mitigation and contingency
R-15 Free resource quota or single database failure blocks service	Medium/High; High priority; Project team; Active	Capacity/cost assumptions and backup/restore experiment; use an approved alternative and record target impacts.
R-16 Speculative patterns obscure mandatory workflow	Medium/Medium; Medium priority; Project team; Active	Compare simple direct coordination with Assignment 2 alternatives; remove unjustified abstraction through reviewed change.
R-17 Evidence path exists but does not prove its claim	Medium/High; High priority; Project team; Active	Human review of evidence content and exact revision; Revert affected acceptance/status and rerun checks

31 Requirements traceability progression

rtm.json is the editable machine-readable matrix; RTM.html is the self-contained searchable live view. Each record contains all 12 required M2 field groups plus approval and acceptance state. The following per-requirement records are embedded here so the PED remains self-contained. The project ownership model is confirmed; it does not establish the presence of detailed architecture diagrams or application artefacts. Empty evidence lists mean no supplied implementation or verification evidence. No Assignment 2 tracking column is added.

FR-01 Not Yet Implemented

Field	Current record
Wording	The system shall allow a citizen to register and log in with a unique account before submitting a request.
Source	Citizens / End Users
Priority	Must Have
Acceptance Criteria	Valid unique registration creates one account and permits login. Duplicate identifiers are rejected. Invalid credentials create no authenticated session; valid credentials access the correct account.
Asr	Candidate driver: access control NFR-02
Architecture	Identity
Data	Account and unique identifier
Design	Registration and authentication contract
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10

Field	Current record
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-02 Not Yet Implemented

Field	Current record
Wording	The system shall allow a citizen to submit a service request or piece of information through a structured form.
Source	Citizens / End Users (Problem & Business Need)
Priority	Must Have
Acceptance Criteria	A valid title, description, location and active category are saved with requester identity and submission time; a unique reference is displayed. Missing or invalid required values create no request. See Section 19.
Asr	Candidate driver: access control NFR-02
Architecture	Request submission
Data	Request, category, requester and reference
Design	Validated submission command
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-03 In Development

Field	Current record
Wording	The system shall allow a requester to view a history of all their submitted requests and each current status.
Source	Citizens / End Users
Priority	Must Have
Acceptance Criteria	Given three own requests in different states, the requester can list and open all three with correct statuses. Another requester cannot access them through the UI or direct requests.
Asr	ASR-L01 atomic history; ASR-L02 access control; ASR-L03 stale-update rejection

Field	Current record
Architecture	RequestLifecycleCoordinator / LifecyclePolicy / SqliteLifecycleStore in product/lifecycle.py
Data	product/schema.sql: accounts, requests, lifecycle_events; status/history transaction and consistent history read
Design	In-process transition and owner-history contracts; ADR-M2-01/02/03 and ADR-M1-02
Technology	Reference slice: Python 3.12.14 / SQLite 3.53.1; ADR-M1-02
Implementation	product/lifecycle.py; product/schema.sql; product/demo.py
Verification	tests/test_lifecycle.py; evidence/verification.txt; evidence/product_demo.json
Status	In Development
References	CR-M2-01; R-10; ADR-M2-03; ADR-M1-02; CR-M2-06; R-12; R-18
Approval	Approved
Acceptance Status	Initial service path verified; full application acceptance not claimed

FR-04 Not Yet Implemented

Field	Current record
Wording	The system shall allow authorised operational staff to view submissions assigned to them or their department.
Source	Operational Staff
Priority	Must Have
Acceptance Criteria	A staff member logged in with the Staff role sees only submissions relevant to their department; unauthorised submissions are not visible or accessible.
Asr	Candidate driver: access control NFR-02
Architecture	Staff work queue
Data	Department and assignment scope
Design	Department-filtered query
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-05 In Development

Field	Current record
Wording	The system shall allow authorised staff to update the status and add remarks to a citizen's request.
Source	Operational Staff
Priority	Must Have
Acceptance Criteria	A permitted update saves status, remark, actor and timestamp. Refreshing within 5 seconds shows the saved state to the requester. Invalid transitions do not alter the record; see FR-13.
Asr	ASR-L01 atomic history; ASR-L02 access control; ASR-L03 stale-update rejection
Architecture	RequestLifecycleCoordinator / LifecyclePolicy / SqliteLifecycleStore in product/lifecycle.py
Data	product/schema.sql: accounts, requests, lifecycle_events; status/history transaction and consistent history read
Design	In-process transition and owner-history contracts; ADR-M2-01/02/03 and ADR-M1-02
Technology	Reference slice: Python 3.12.14 / SQLite 3.53.1; ADR-M1-02
Implementation	product/lifecycle.py; product/schema.sql; product/demo.py
Verification	tests/test_lifecycle.py; evidence/verification.txt; evidence/product_demo.json
Status	In Development
References	CR-M2-01; R-10; ADR-M2-01; ADR-M2-02; ADR-M1-02; CR-M2-06; R-12; R-18
Approval	Approved
Acceptance Status	Initial service path verified; full application acceptance not claimed

FR-06 Not Yet Implemented

Field	Current record
Wording	The system shall allow administrators to create, edit and deactivate accounts and assign roles, with updated access enforced on subsequent authenticated requests.
Source	System Administrators
Priority	Must Have
Acceptance Criteria	An administrator can create an account, edit permitted details, deactivate it and assign/change roles. Deactivation and role changes take effect by the next authenticated request. Non-administrators cannot invoke these operations.
Asr	Candidate driver: access control NFR-02
Architecture	Administration and identity
Data	Account activation and role membership
Design	Recheck current access on every request

Field	Current record
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-07 Not Yet Implemented

Field	Current record
Wording	The system shall validate mandatory fields and reject a request submission if required information is missing or malformed.
Source	Citizens / End Users; Municipality / Organisation
Priority	Must Have
Acceptance Criteria	Missing or invalid title, description, location or category values are rejected with field-level errors. The same rules apply to direct submissions. No invalid record is saved; limits are defined in Section 19.
Asr	Candidate driver: access control NFR-02
Architecture	Submission validation
Data	Required field and category integrity
Design	Shared boundary validation policy
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-08 In Development

Field	Current record
Wording	The system shall give requesters meaningful feedback when a request is accepted, rejected, updated or completed.
Source	Master brief Section 3 requester capabilities

Field	Current record
Priority	Must Have
Acceptance Criteria	Within 60 seconds of a saved lifecycle event, authenticated request history/details show status, event time and a requester-visible explanation. Rejection and resolution require explanations. Email/push transport remains optional.
Asr	ASR-L01 atomic history; ASR-L02 access control; ASR-L03 stale-update rejection
Architecture	RequestLifecycleCoordinator / LifecyclePolicy / SqliteLifecycleStore in product/lifecycle.py
Data	product/schema.sql: accounts, requests, lifecycle_events; status/history transaction and consistent history read
Design	In-process transition and owner-history contracts; ADR-M2-01/02/03 and ADR-M1-02
Technology	Reference slice: Python 3.12.14 / SQLite 3.53.1; ADR-M1-02
Implementation	product/lifecycle.py; product/schema.sql; product/demo.py
Verification	tests/test_lifecycle.py; evidence/verification.txt; evidence/product_demo.json
Status	In Development
References	CR-M2-01; R-10; ADR-M2-03; ADR-M1-02; CR-M2-06; R-12; R-18
Approval	Approved
Acceptance Status	Initial service path verified; full application acceptance not claimed

FR-09 Not Yet Implemented

Field	Current record
Wording	The system shall provide authorised management users with basic service activity and accountability information.
Source	Master brief Section 3 management capabilities
Priority	Must Have
Acceptance Criteria	For a known dataset and selected submission-date range, counts and lists correctly identify open, overdue, resolved and closed requests by category/status, with assignee and age visible. Unauthorised access is denied. See Section 19 definitions.
Asr	Candidate driver: access control NFR-02
Architecture	Management reporting
Data	Date, category, assignee, status and due time
Design	Authorised aggregate and detail query
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented

Field	Current record
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-10 Not Yet Implemented

Field	Current record
Wording	The system shall categorise requests using an active controlled category.
Source	Master brief Section 3 requester capabilities
Priority	Must Have
Acceptance Criteria	Unknown/inactive categories are rejected. Administrators maintain the list; retirement preserves historical request category labels.
Asr	Candidate driver: access control NFR-02
Architecture	Category administration
Data	Stable category ID and historical label
Design	Retire categories without deleting history
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-11 Not Yet Implemented

Field	Current record
Wording	The system shall let authorised staff search, filter and sort relevant requests and view full details.
Source	Master brief Section 3 staff capabilities
Priority	Must Have
Acceptance Criteria	Reference/title search and category/status/assignee filters return matching authorised records. Submitted-date/due-date sorting is correct. Details show description, location, owner, times and lifecycle actions; inaccessible records remain hidden.
Asr	Candidate driver: access control NFR-02
Architecture	Staff request query

Field	Current record
Data	Scoped searchable request fields
Design	Scope before search filter and sort
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

FR-12 In Development

Field	Current record
Wording	The system shall let authorised staff assign or accept responsibility for a request.
Source	Master brief Section 3 staff capabilities
Priority	Must Have
Acceptance Criteria	Staff can accept unassigned work or assign an eligible staff member. Assignee, actor and time are recorded. Ineligible/unauthorised assignments are rejected without changing ownership.
Asr	ASR-L01 atomic history; ASR-L02 access control; ASR-L03 stale-update rejection
Architecture	RequestLifecycleCoordinator / LifecyclePolicy / SqliteLifecycleStore in product/lifecycle.py
Data	product/schema.sql: accounts, requests, lifecycle_events; status/history transaction and consistent history read
Design	In-process transition and owner-history contracts; ADR-M2-01/02/03 and ADR-M1-02
Technology	Reference slice: Python 3.12.14 / SQLite 3.53.1; ADR-M1-02
Implementation	product/lifecycle.py; product/schema.sql; product/demo.py
Verification	tests/test_lifecycle.py; evidence/verification.txt; evidence/product_demo.json
Status	In Development
References	CR-M2-01; R-10; ADR-M2-02; ADR-M1-02; CR-M2-06; R-12; R-18
Approval	Approved
Acceptance Status	Initial service path verified; full application acceptance not claimed

FR-13 In Development

Field	Current record
Wording	The system shall enforce controlled status transitions and record actions and resolution information.
Source	Master brief Section 3 staff capabilities
Priority	Must Have
Acceptance Criteria	Permit Submitted to Accepted or Rejected; Accepted to In Progress; In Progress to Resolved; Resolved to Closed. Only authorised staff transition records. Acceptance requires an assignee and due time; rejection requires a reason and resolution requires a summary. Record actor/time/explanation. Reject all other transitions.
Asr	ASR-L01 atomic history; ASR-L02 access control; ASR-L03 stale-update rejection
Architecture	RequestLifecycleCoordinator / LifecyclePolicy / SqliteLifecycleStore in product/lifecycle.py
Data	product/schema.sql: accounts, requests, lifecycle_events; status/history transaction and consistent history read
Design	In-process transition and owner-history contracts; ADR-M2-01/02/03 and ADR-M1-02
Technology	Reference slice: Python 3.12.14 / SQLite 3.53.1; ADR-M1-02
Implementation	product/lifecycle.py; product/schema.sql; product/demo.py
Verification	tests/test_lifecycle.py; evidence/verification.txt; evidence/product_demo.json
Status	In Development
References	CR-M2-01; R-10; ADR-M2-01; ADR-M2-02; ADR-M1-02; CR-M2-06; R-12; R-18
Approval	Approved
Acceptance Status	Initial service path verified; full application acceptance not claimed

NFR-01 Not Yet Implemented

Field	Current record
Wording	Usability: The citizen-facing submission process shall be simple enough for a first-time user to complete without assistance.
Source	Citizens / End Users (Possible Stakeholder Conflict)
Priority	Must Have
Acceptance Criteria	With five first-time representative users and the same valid scenario, at least four submit successfully within 5 minutes without assistance. Record individual times, errors and feedback.
Asr	Candidate driver: NFR-01
Architecture	Citizen submission UI
Data	Minimal required input

Field	Current record
Design	Accessible form and usability scenario
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

NFR-02 In Development

Field	Current record
Wording	Security: Access to citizen personal information and request data shall be restricted according to the authenticated user's role.
Source	System Administrators; Municipality / Organisation
Priority	Must Have
Acceptance Criteria	A user with the Citizen role cannot view, via the UI or a direct request, another citizen's submissions or personal data; role-based access is enforced on every relevant request.
Asr	ASR-L01 atomic history; ASR-L02 access control; ASR-L03 stale-update rejection
Architecture	RequestLifecycleCoordinator / LifecyclePolicy / SqliteLifecycleStore in product/lifecycle.py
Data	product/schema.sql: accounts, requests, lifecycle_events; status/history transaction and consistent history read
Design	In-process transition and owner-history contracts; ADR-M2-01/02/03 and ADR-M1-02
Technology	Reference slice: Python 3.12.14 / SQLite 3.53.1; ADR-M1-02
Implementation	product/lifecycle.py; product/schema.sql; product/demo.py
Verification	tests/test_lifecycle.py; evidence/verification.txt; evidence/product_demo.json
Status	In Development
References	CR-M2-01; R-10; ADR-M2-03; ADR-M1-02; CR-M2-06; R-12; R-18
Approval	Approved
Acceptance Status	Initial service path verified; full application acceptance not claimed

NFR-03 Not Yet Implemented

Field	Current record
Wording	Performance: Status lookup shall meet the defined response-time target under the specified test load.

Field	Current record
Source	Citizens / End Users; Operational Staff
Priority	Should Have
Acceptance Criteria	With 50 concurrent authenticated users and 10,000 requests, each user performs one lookup every 5 seconds for 10 minutes after warm-up. At least 95% complete within 3 seconds; fewer than 1% fail. Record hardware/network conditions.
Asr	Candidate driver: NFR-03
Architecture	Request lookup and persistence
Data	Lookup indexes and representative dataset
Design	Repeatable load scenario
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

NFR-04 Not Yet Implemented

Field	Current record
Wording	Availability: The platform shall be available for citizen use during standard business hours with minimal unplanned downtime.
Source	Municipality / Organisation; Project Client / Representative
Priority	Should Have
Acceptance Criteria	Over 20 weekdays, achieve at least 99% availability during 08:00–17:00 Africa/Johannesburg, measured by one-minute probes. Exclude only maintenance announced 24 hours ahead and report excluded minutes separately.
Asr	Candidate driver: NFR-04
Architecture	Hosting and operations
Data	Durable storage and recovery
Design	Availability measurement and restore procedure
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented

Field	Current record
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

NFR-05 Not Yet Implemented

Field	Current record
Wording	Data Quality: The system shall enforce validation rules on all citizen-submitted data to reduce incomplete or malformed records.
Source	Operational Staff; Municipality / Organisation
Priority	Should Have
Acceptance Criteria	All valid/invalid boundary cases for submission fields pass validation tests. An audit of 100 saved requests (or all if fewer) finds no missing mandatory or invalid-category values.
Asr	Candidate driver: NFR-05
Architecture	Submission policy and persistence
Data	Mandatory fields and category references
Design	Boundary validation and integrity checks
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

NFR-06 In Development

Field	Current record
Wording	Maintainability: The system's requirements, decisions and changes shall be traceable to support controlled maintenance after M1.
Source	Project Client / Representative (SEN381 Master Project Brief)
Priority	Should Have
Acceptance Criteria	Each baselined requirement can be traced, via the RTM, to its source, its acceptance criteria and any linked design or test evidence added in later milestones.
Asr	ASR-T01 traceability
Architecture	Implemented local traceability control module

Field	Current record
Data	Versioned JSON requirements, exact approved-change records and SHA-256 evidence manifest
Design	ADR-M2-04 in ADRs.md; validate, diff, verify-manifest and render command interfaces; ADR-M1-01
Technology	ADR-M1-01 Python standard library tooling
Implementation	traceability.py
Verification	tests/test_traceability.py; evidence/verification.txt
Status	In Development
References	ADR-M1-01; CR-M2-01; R-04; R-10; CR-M2-05; R-17; ADR-M2-04
Approval	Approved
Acceptance Status	Initial automated tooling checks passed; full lifecycle traceability continues

NFR-07 Not Yet Implemented

Field	Current record
Wording	Scalability: The design shall support defined growth in users and requests while preserving lookup behaviour.
Source	Municipality / Organisation
Priority	Could Have
Acceptance Criteria	At M2 assess 50,000 stored requests and 100 concurrent users against the same lookup pattern and 3-second 95th-percentile target. Record capacity assumptions and bottlenecks; load-test at M3. Resource/configuration increases are permitted with cost recorded.
Asr	Candidate driver: NFR-07
Architecture	Architecture and persistence capacity
Data	50000 requests and query access patterns
Design	Capacity assessment before load testing
Technology	Technology stack record; see consolidated handover
Implementation	Not Yet Implemented
Verification	Not Yet Implemented
Status	Not Yet Implemented
References	CR-M2-01; R-10
Approval	Approved
Acceptance Status	Product acceptance not demonstrated

32 Traceability control decision and implementation

ADR-M1-01 tooling direction is confirmed through the approved v2.0 approval. Version 2.0.1 adds exact change-record and evidence-hash checks under CR-M2-05. Problem: 20 stable requirements now need evidence across architecture, data, design, technology, implementation and tests, while human approval must remain distinct from automated checks. ASR-T01 derives from NFR-06: retain every M1 ID/source/criterion, detect missing fields and broken local evidence, and prevent an Implemented status with no implementation or verification links.

Decision: use versioned UTF-8 JSON, a Python standard-library command-line validator, baseline comparison and a generated read-only HTML view. Alternatives: DOCX-only tables are easy to edit but harder to validate consistently; a database or hosted requirements platform adds setup and access dependencies beyond this small register. JSON introduces synchronisation work: edit the JSON and regenerate the view, then review the PED snapshot in the same change. The tool checks structural integrity, not evidence truth or human identity.

Architecture responsibility is a local engineering-support module, not the deployed citizen application. Data decision is one record per stable ID with source and criteria preserved; no citizen data or secrets are stored. Interface is validate(root) plus CLI exit 0 for valid structure, 1 for invalid structure. Paths must remain inside the evidence pack; traversal and external paths are rejected. Unreferenced baseline differences fail validation; referenced changes still require human review of the change record. Technology decision applies only to tooling, not the application stack.

Complete tooling trace: NFR-06 -> ASR-T01 and M1 quality constraint -> traceability.validate -> rtm.json and baseline_m1.json -> explicit validation and status checks -> ADR-M1-01 -> traceability.py -> tests/test_traceability.py and evidence/verification.txt. This proves an initial traceability-support implementation. It does not replace the M2 requirement to demonstrate a meaningful CivicConnect product path.

33 Product path integration record

Executed product path: FR-13 with FR-05 and FR-08. Active staff accept a Submitted request with a valid assignee and due time; status and history commit atomically; the owner reads the committed status and explanation through a separate connection. Product_Trace.md and product_trace.json identify every stage and its concrete evidence.

Trace stage	Expected evidence and owner
Requirement	FR-13 controls transitions; FR-05 saves remarks, actor and time; FR-08 exposes meaningful requester feedback. Evidence: baseline_m1.json
ASR and constraint	ASR-L01 mandatory status/history atomicity, ASR-L02 ownership and staff-department access, ASR-L03 stale-update rejection. NFR-02 and the M1 no-budget constraint apply. Evidence: Product_Trace.md

Trace stage	Expected evidence and owner
Architecture responsibility	RequestLifecycleCoordinator receives an in-process command, LifecyclePolicy checks the transition, and SqliteLifecycleStore owns atomic persistence and authorised history reads. Evidence: product/lifecycle.py
Data decision	Accounts, requests and lifecycle_events; unique reference/version, foreign keys, status constraints, one write transaction and a consistent read snapshot. Evidence: product/schema.sql
Design and interface	transition(actor_id, reference, target, expected_version, explanation, assignee, due_at); history(actor_id, reference). Explicit Forbidden, Conflict and InvalidTransition failures. Evidence: product/lifecycle.py
Technology and ADR	ADR-M1-02 uses Python 3.12.14 and bundled SQLite 3.53.1 for this executable reference slice. ADR-M2-01/02/03 supply the approved domain decisions. Evidence: ADRs.md
Application artefact	The real service commits Submitted to Accepted, then a separate connection reads the same status and explanation for the owning citizen. Evidence: product/demo.py
Initial verification	20 lifecycle/persistence/access tests plus 15 traceability tests pass. Demo output records the saved event and an actual local post-commit read time. Evidence: evidence/verification.txt

Initial trace verification is complete: 20 product tests and 15 traceability tests pass. The product demo records the actual saved status/history and local read duration. The full browser authentication/UI, deployed response targets and production capacity remain outside this initial slice; the RTM records In Development rather than full acceptance.

34 Human verification and approval register

HV-01 to HV-16 and the completion update are approved. The project team have confirmed the document approval. M1 sign-off date: 09/09/2026. M2 sign-off date: 30/09/2026. Approval_Record.md preserves the document identity; the GitHub review record is maintained separately.

Review ID and coverage	Required record	State
HV-01 Problem and stakeholder needs	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-02 Scope inclusions exclusions and deferrals	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-03 FR-01 through FR-13 wording source priority and criteria	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved

Review ID and coverage	Required record	State
HV-04 NFR-01 through NFR-07 targets and feasibility	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-05 RTM coverage and source links	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-06 Constraints and team registration	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-07 Risks R-01 through R-09	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-08 All seven forward engineering considerations	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-09 EDL-001 through EDL-007	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-10 GitHub settings tasks branches PR and two approvals	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-11 AI-001 through AI-003 human verification scope	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-12 Document control version history and baseline identity	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-13 Working agreement and four listed identities	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-14 Section 19 approved definitions and quality targets	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-15 M1 final visual review and team baseline sign-off	Approved. M1 review date 09/09/2026; team confirmation recorded 24/09/2026.	Approved
HV-16 M2 ADRs risks RTM tool and product trace	M2 document and completion update: Approved. Sign-off date: 30/09/2026. Section 38 and the retained logs provide initial product-path evidence.	Approved; initial trace verified
Reviewer	M2 approval	Date revision and evidence
Marco Brazao de Sousa 601587	Approved	30/09/2026; Approved
Gift Mathe 600731	Approved	30/09/2026; Approved
Masana Sakala 600688	Approved	30/09/2026; Approved
Jacobus Petrus Geldenhuys 602755	Approved	30/09/2026; Approved

AI-M2-001 is recorded in the uniform project AI Usage Register in Section 16. The related integration, traceability and test evidence is retained with the project artefacts.

35 Delivery and Presentation

Responsibility	Evidence and completion limit
PED continuity and control	Same source PED extended to v2.0.2; historical content and approved v2.0 source preserved. Documentation approved.
Baseline review and change control	Sections 27 and 28; no silent requirement changes.
Risks assumptions and forward concerns	Sections 29 and 30; project-wide risk ownership is recorded and mitigations remain active.
RTM status and completeness	All 20 requirements and all required field groups, JSON plus live HTML and embedded records.
Trace and consistency	Executed product and tooling traces; M1/A2 decisions, actual code, RTM and evidence aligned. Section 38 and Product_Trace.md.
Lifecycle implementation	The project team delivered the working lifecycle service, schema, demonstration and validator with 35 tests. Issue #8 and PR #9 are recorded in Governance.md. Independent reviews are recorded separately.
Presentation	Presentation.md gives demonstration route, talking points, defence questions and evidence locations.

The approved PED now includes the executed initial product path. Later M2 application artefacts will be reconciled to these requirement IDs through the same controlled-change process. Independent reviews remain part of the GitHub merge process.

36 Completion Audit

The document is approved. The initial product path is implemented and verified, and all 20 baseline requirements are unchanged. The following audit summarises the deliverables and supporting evidence.

Task	State	Evidence and remaining action
1 PED continuity	Complete	v2.0.2 continues the same PED; previous approved source and hashes retained.
2 M1 review	Complete	All 20 original requirements, sources, priorities and criteria unchanged; exact diff is empty.
3 Controlled changes	Complete	CR-M2-01 to CR-M2-06; exact approved-change checking and preserved source history.
4 Risks assumptions and forward concerns	Complete for this contribution	Complete for this milestone record

Task	State	Evidence and remaining action
5 RTM progression	Complete	All required fields for 20 requirements; real lifecycle and tooling evidence attached.
6 Evidence-based statuses	Complete	Initial path is In Development; unrelated features remain Not Yet Implemented. Approval is separate.
7 End-to-end product trace	Complete for initial service path	FR-13/05/08 -> ASR -> architecture -> data -> interface -> technology/ADR -> actual service -> 20 product tests and executed demo.
8 PED consistency	Available evidence reconciled; final integration outstanding	Project records reconciled to ADR-M2-01/02/03/04, source and RTM. Final architecture, data and technology records are reconciled through the same project review process.
9 Development and GitHub	Implemented and published for review	Real branch/commits; 35 tests; issue #8 and PR #9 published; links retained in Governance.md. Independent reviews are performed by the reviewers.
10 Presentation	Preparation complete	Timed script, live commands, source navigation and defence answers. The assessed presentation follows the prepared script.

Verification and change impact

CR-M2-05 changes document control and tooling only. No scope, requirement text, priority, source or acceptance target changes. The validator now requires an Approved CR record identifying the exact old/new field values and a local approval-evidence file before permitting a baseline edit. The manifest command detects changed evidence bytes. Tests include bogus CR labels, mismatched approved changes, exact diff output and altered evidence. Actual results and commands are in evidence/verification.txt. These automated checks do not authenticate people or prove product behaviour.

AI-M2-002 is recorded in the uniform project AI Usage Register in Section 16. Verification results are retained in evidence/verification.txt; independent GitHub reviews remain part of the normal merge process.

37 Approved research to decision integration

The following CivicConnect decisions based on the M1 baseline and Assignment 2 are approved. They are applied in the initial service path described in Section 38 and the RTM.

ADR-M2-01 Lifecycle coordination

Status: Approved. Source: FR-05 and FR-13; A2 Task 1A

Use a focused RequestLifecycleCoordinator to authorise, validate and coordinate the status/history transaction. Keep lifecycle rules in LifecyclePolicy and persistence behind its contract.

Alternatives considered: Observer was rejected for mandatory status/history work because registration and ordering obscure required steps. A generic mediator framework is unnecessary.

Consequences: Mandatory work is explicit and testable. The coordinator must remain focused; new required steps change orchestration.

ADR-M2-02 Persistence integrity

Status: Approved. Source: FR-05, FR-12 and FR-13; A2 Task 2

Enforce authorisation and lifecycle rules in the application; enforce structural constraints in persistence. Commit status and history in one transaction. Use an atomic expected-version check and revalidation after a conflict.

Alternatives considered: Application-only validation cannot protect structural integrity from every write path. Blind retries and independent history writes were rejected.

Consequences: Failures leave neither a partial status update nor a partial history event. Conflict handling adds explicit failure paths and tests.

ADR-M2-03 Feedback and internal interface

Status: Approved. Source: FR-03, FR-08 and NFR-02; A2 Task 1B and Task 3

Read requester-visible feedback from committed lifecycle history under ownership checks. Use an in-process application/service contract for internal calls. Introduce REST only at a real client/server or external boundary.

Alternatives considered: A multi-channel Strategy hierarchy and provider Adapter are unnecessary for the current in-app scope. An internal network split adds deployment and failure complexity without a current requirement.

Consequences: Feedback stays consistent with committed state. Additional delivery channels require controlled scope and interface changes.

ADR-M2-04 Traceability and review integration

Status: Approved. Source: NFR-06; A2 Task 4

Keep short-lived branches, substantive PRs, two independent approvals and repeatable local checks. Store RTM and exact change records in versioned JSON and generate the live HTML from that data.

Alternatives considered: Direct writes to main and document-only evidence checks were rejected. A hosted requirements database adds unnecessary operational overhead for 20 requirements.

Consequences: Checks expose broken links and unauthorised baseline differences. Human review still assesses correctness and provides independent approvals.

38 Executed initial lifecycle trace

The initial application path demonstrates the approved lifecycle, integrity and feedback decisions. Product_Trace.md records the complete trace, with product_trace.json providing its machine-readable form.

Requirement

FR-13 controls transitions; FR-05 saves remarks, actor and time; FR-08 exposes meaningful requester feedback. Evidence: baseline_m1.json

ASR and constraint

ASR-L01 mandatory status/history atomicity, ASR-L02 ownership and staff-department access, ASR-L03 stale-update rejection. NFR-02 and the M1 no-budget constraint apply. Evidence: Product_Trace.md

Architecture responsibility

RequestLifecycleCoordinator receives an in-process command, LifecyclePolicy checks the transition, and SqliteLifecycleStore owns atomic persistence and authorised history reads. Evidence: product/lifecycle.py

Data decision

Accounts, requests and lifecycle_events; unique reference/version, foreign keys, status constraints, one write transaction and a consistent read snapshot. Evidence: product/schema.sql

Design and interface

transition(actor_id, reference, target, expected_version, explanation, assignee, due_at); history(actor_id, reference). Explicit Forbidden, Conflict and InvalidTransition failures. Evidence: product/lifecycle.py

Technology and ADR

ADR-M1-02 uses Python 3.12.14 and bundled SQLite 3.53.1 for this executable reference slice. ADR-M2-01/02/03 supply the approved domain decisions. Evidence: ADRs.md

Application artefact

The real service commits Submitted to Accepted, then a separate connection reads the same status and explanation for the owning citizen. Evidence: product/demo.py

Initial verification

20 lifecycle/persistence/access tests plus 15 traceability tests pass. Demo output records the saved event and an actual local post-commit read time. Evidence: evidence/verification.txt

Verification result and practical scope

Verification passed: 20 lifecycle tests and 15 traceability tests. The executed demo confirms atomic status and history persistence followed by a separate-connection owner history read. Retained results are recorded in evidence/verification.txt and evidence/product_demo.json.

This section records the initial M2 lifecycle reference slice. The integrated M3 application adds the web interface, login/session handling, citizen request flows, staff queues, management reporting and account controls. M3 implementation and verification evidence is recorded in Sections 39 to 44 and the linked project evidence.

Reference slice decision and risk

ADR-M1-02 approves Python 3.12.14 and SQLite 3.53.1 for the executable reference slice. In-memory mocks were rejected because they cannot prove database commit/rollback. A web framework and hosted database were rejected for this initial path because they would impose wider stack and deployment choices. Full application selection remains Technology work.

R-18: treating the reference slice as a production-ready application. Probability Medium; impact High; priority High; owner Traceability and technology leads. Mitigation: label its authentication, UI, deployment and load limits, preserve In Development statuses and test any replacement application separately. Contingency: correct overclaims and reopen affected acceptance records. Status Active.

CR-M2-06 adds initial implementation evidence without altering requirement wording, priorities, sources or criteria. The approved baseline is preserved. A2 Task 1A supplies coordinator rationale, Task 1B feedback scope, Task 2 transaction/versioning integrity and Task 3 in-process boundaries. The implementation is reconciled to those decisions in the RTM.

AI-M2-003 is recorded in the uniform project AI Usage Register in Section 16. The reference-slice test output is retained with the project evidence, and independent pull-request reviews are recorded separately.

39 M3 version control

PED v3.0 continues the same controlled document from the approved v2.0.2 baseline. Sections 39 to 44 record the integrated M3 application, local demonstration scope, verification, team-reviewed registers and release decision. Earlier sections preserve the approved baseline and are read together with these M3 updates.

Item	Record
Version	PED v3.0, team-reviewed release document. Previous version: v2.0.2, SHA-256 a745a93cf25f46007cdff0c38ce34781478b0772ecc5193d7c9cdd2022ac8df4.
Date	16 October 2026; team verification and required approvals completed.
Release candidate assessed	Integrated M3 application: main commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659. The separate lifecycle-slice report retains RC-M3-candidate-8bf449ede114 and its own fingerprint.
Earlier draft context	The earlier draft predated the team's integrated M3 evidence review. This final PED incorporates the team-reviewed application, CI, test and approval records.
Change control	This documentation update records the team verification and local-release decision; it does not alter M3 application code.

40 Staging environment and critical journey

The approved release environment is a local run of the integrated M3 application described in its project README. The separate staging scripts under staging/ exercise a simulated copy of the three-file lifecycle-service slice; they are supporting evidence for that slice, not a deployment of the integrated web app. All demonstration records use synthetic data.

The integrated M3 application revision reviewed by the team is main commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659. GitHub Actions run 38870644636 completed successfully on that commit after PR #16 merged. The lifecycle-slice staging report retains its own release id RC-M3-candidate-8bf449ede114 and fingerprint; that identity must not be confused with the M3 application commit.

40.1 Verification results (13 of 13 pass)

ID	Check	Requirement	Result
ST-01	Release files match deployed manifest and service loads from release dir	NFR-06	PASS
ST-02	Staging database schema and integrity	NFR-06 / R-12	PASS
ST-03	Critical journey: accept, progress, resolve, close; citizen reads status/history	FR-13, FR-05, FR-08	PASS
ST-04	Committed state survives process restart	FR-05 / R-12	PASS
ST-05	Citizen B cannot read citizen A's history	NFR-02	PASS
ST-06	Wrong-department staff cannot transition	NFR-02	PASS
ST-07	Citizen cannot transition a request	NFR-02	PASS
ST-08	Deactivated staff account is denied	NFR-02 / FR-06	PASS
ST-09	Invalid transition from Closed is rejected	FR-13	PASS

ID	Check	Requirement	Result
ST-10	Stale version cannot overwrite	FR-13 / ASR-L03	PASS
ST-11	Rejection without reason is refused and leaves no record	FR-13, FR-08	PASS
ST-12	Rejection with reason is committed and visible	FR-13, FR-08	PASS
ST-13	Backup and restore reproduces committed data (R-15 experiment)	R-15 / NFR-04	PASS

The integrated M3 application implements the citizen, staff and management paths described in the product README. The recorded local test suite reports 37 tests passed, with no failures or errors. The project team confirmed that the planned requirements, tests, checks and evidence were tested, checked and verified. The shared CI workflow passed on the identified main commit. The 13-of-13 staging checks in evidence/staging_verification.json cover the separate lifecycle-service slice.

The separate lifecycle-slice robustness probe tried 13 malformed inputs: 10 returned documented domain errors and 3 produced a raw sqlite3.ProgrammingError. This finding is retained as a low-severity limitation of that probe; it is not represented as an integrated web-application test result. See evidence/robustness_probe.json and the Defect Register.

41 Performance test

The team reran and reviewed the 10-minute NFR-03 profile for the separate lifecycle-service slice on 16 October. It recorded 6,000 lookups, zero failures and a p95 of 0.825 ms. The generator and service shared one Linux host and process with local SQLite; the run did not measure the integrated web request path. The team verified the planned M3 performance checks for the local school-project scope; this profile retains its stated workload and evidence boundary.

Item	Record
Workload	50 virtual users, 10,000 stored requests (1,000 citizens, 20 staff, 0 to 4 events per request), one authorised lookup per user every 5 seconds, 60 seconds warm-up discarded, 600 seconds measured. Each lookup opened its own connection.
Environment	Linux sandbox, 1 CPU core, 4 GB RAM, Python 3.12.3, SQLite 3.45.1, local file database, load generator and service in one process.
Samples and failures	6,000 samples, 0 failures (target: below 1%)
Latency	median 0.49 ms; p95 0.82 ms (target 3 s); p99 1.25 ms; maximum 8.18 ms; share within 3 s 100% (target at least 95%)
Supplementary headroom run	50 users, no think time, 30 s after 5 s warm-up: 131,197 lookups, 0 failures, about 4,373 per second, p95 26.60 ms, p99 72.20 ms, maximum 221.75 ms. Not a capacity claim.
Limitations	This supplemental read-only profile covers 10,000 requests and 50 virtual users; it is not the separate 50,000-request/100-user scenario. The team verified the planned M3 performance checks for the local school-project scope. No HTTP, TLS, authentication, rendering, network or write-contention performance claim is made.
Conclusion	NFR-03 is met at the data layer under the specified profile. It is not demonstrated for a deployed application.
Evidence	evidence/perf_nfr03_profile.json; evidence/perf_stress_supplementary.json; staging/load_test.py

42 M3 registers

The M3 risk, defect and technical-debt registers were reviewed by the team on 16 October 2026. Their standalone copies are in registers/. The single AI-use record is maintained in Section 16 and synchronized to the repository copy. The registers record the approved local-release scope, completed team verification and follow-up items outside that scope.

42.1 Risk Register updates and new risks

Updates to existing risks:

ID	Risk	Current status	Evidence / treatment
R-08	Citizen data exposure	Reduced for exercised local paths	M3 login, sessions and role/ownership controls; web-security tests and CI passed.
R-09	Hosting access or cost	Outside local project scope	The school-project release runs locally; no hosting account is required.
R-12	Partial lifecycle save or stale update	Reduced for tested paths	Transactional tests, integration tests and successful CI.
R-13	Runtime incompatibility	Reduced	CI passed on configured Python 3.12; local result used Python 3.15.
R-15	Data loss or database failure	Recorded follow-up	Lifecycle slice backup/restore passed; hosted operations are outside local scope.
R-18	Reference slice mistaken for full app	Mitigated in this document set	PED and evidence summary distinguish M3 integrated app from slice experiment.

New risks:

ID	Risk and cause	Prob.	Impact	Priority	Mitigation	Contingency	Owner	Status
R-19	Five-person first-time-user study for NFR-01 was deferred while the load-reliability issue was prioritised; insufficient time remained before M4 closure.	Medium	High	High	Team-reviewed reason and unmet status recorded; retain as Must Have closure item.	Complete the defined study or secure a formally approved baseline change.	Team	Open; NFR-01 unmet
R-20	NFR-04 hosted availability has no 20-weekday one-minute probe evidence.	Low for local demo	High for production	Medium	State the local-only boundary; do not make a hosted availability claim.	Collect the defined probe evidence if hosted release enters scope.	Team	Open; NFR-04 unmet

ID	Risk and cause	Prob.	Impact	Priority	Mitigation	Contingency	Owner	Status
R-21	Integrated website failed the NFR-03 request reliability threshold: 647 of 6,000 requests failed (10.78333%).	Medium	High	High	Production no-go; retain exact load evidence and acceptance threshold. Root cause is not confirmed.	Diagnose resets, correct the issue and repeat the 50-user profile.	Team	Open; production blocker
R-22	Lifecycle staging evidence is a separate service slice, not the integrated website.	Low	Medium	Medium	Keep evidence scope labelled and use the integrated website results for M4 release decisions.	Correct any evidence attribution that crosses the boundary.	Team	Mitigated; boundary retained
R-23	Application commit, CI run and lifecycle-slice fingerprints may be confused.	Low	Medium	Medium	Identify the assessed app/evidence commit as c8fcc9887d58bc07e7b592e292dbe05222d97190 and label slice identifiers separately.	Correct any mismatched record before submission.	Team	Mitigated; identifiers disambiguated
R-24	The local M4 document set may diverge from protected GitHub main before merge.	Medium	High	High	Submit the controlled M4 documentation PR, complete required non-author review and record the resulting merge SHA.	Keep the final package and SHA pending until merge is confirmed.	Team	Open until PR merge
R-25	Production release could proceed before reliability and availability evidence is sufficient.	Low	High	High	Document the M4 NO-GO decision and limit the demo to synthetic data on local loopback.	Do not authorise production pending all release-reconsideration criteria.	Team	Mitigated by production NO-GO

42.2 Defect Register

ID	Finding	Severity	Scope	Found by	Status	Evidence	Disposition
DEF-001	PED heading 1.5 should be 10.5.	Low	PED baseline	Document review	Fixed in v3.0	Section 10.5	v2.0 baseline preserved
DEF-002	Mojibake dash report in a README.	Low	Legacy pack report	Text review	Not reproduced in current root README or inspected M2 pack	README review	Not an M3 release defect

ID	Finding	Severity	Scope	Found by	Status	Evidence	Disposition
DEF-003	List reference or assignee raised raw SQLite error in malformed-input probe.	Low	Lifecycle-slice probe	Robustness probe	Retained for that slice	evidence/robustness_probe.json	Does not describe an integrated web-path result
DEF-004	Integrated CivicConnect website recorded 647 failed requests out of 6,000 (10.78333%) under the 50-user load profile; p95 latency was 1,051.639 ms.	High for production readiness	Integrated local website; synthetic data; loopback HTTP	M4 website load profile reviewed by the team	Open; production no-go; root cause unconfirmed	evidence/website-stage-20261016/ staging_load_report.json; staging_load_report.md; load_samples.csv	Diagnose the connection resets and repeat the profile without weakening the M1 threshold.

The integrated M3 local test record reports 37 passed, 0 failed and 0 errors; the main-branch CI run also passed. The low-severity findings below are retained with their original scope and do not prevent the approved local school-project release.

42.3 Technical Debt Register

ID	Debt / limitation	Evidence	Follow-up	Priority	Owner	Status	Local release effect
TD-001	Initial slice lacked authentication and sessions.	M3 serve.py/web.py and security tests	Keep local credentials private.	Low	M1	Addressed	None
TD-002	Initial slice lacked a web interface.	M3 server-rendered app	None for current scope.	Low	M1	Addressed	None
TD-003	Initial slice lacked staff/management views.	M3 queues and reports	None for current scope.	Low	M1	Addressed	None
TD-004	No general schema migration framework.	schema.sql and initialization	Add if schema changes require it.	Medium	M1	Open	None for current demo
TD-005	SQLite is a single local database.	Lifecycle implementation	Reassess only if scale changes.	Low	M1/M4	Accepted	Suitable for current scope
TD-006	No operational monitoring service.	Local run model	No action for local project.	Low	M1	Accepted	None
TD-007	Settings are supplied as process environment variables.	.env.example and serve.py	Follow README and keep secrets local.	Low	M3	Documented	None

ID	Debt / limitation	Evidence	Follow-up	Priority	Owner	Status	Local release effect
TD-008	Local histories used different Python versions.	Python 3.15 local and Python 3.12 CI	Use Python 3.12 documented baseline.	Low	M3	Recorded	None
TD-009	Request history has no pagination.	Current local app	Add only if data size requires it.	Low	M1	Accepted	None
TD-010	PED and RTM evidence require review when code changes.	PED v3.0 and evidence files	Keep linked records synchronized.	Low	M2/M3	Addressed for this release	None
TD-011	The staging harness simulates a lifecycle slice.	staging/README_Staging.md	Use product README for M3 demo.	Low	M4	Accepted for local scope	None
TD-012	Synchronized 50-user integrated-web reliability criterion failed; root cause remains unconfirmed.	647/6000 failed (10.78333%); p95 1,051.639 ms.	Diagnose resets and repeat the documented profile with unchanged thresholds.	High for release readiness	Team	Open; production blocker	Production NO-GO
TD-013	Five-person first-time-user usability study was deferred while the load issue was prioritised; insufficient time remained before M4 closure.	NFR-01 team-reviewed closure status and M1 acceptance criterion.	Complete the defined study or obtain a formally approved M1 baseline change.	High for acceptance closure	Team	Open; Must Have unmet	Usability not demonstrated
TD-014	No hosted 20-weekday, one-minute availability probe is recorded.	NFR-04 acceptance criterion; no qualifying probe record.	Collect the defined evidence if hosted operation enters scope.	Medium	Team	Open; production evidence absent	No hosted availability claim

42.4 AI register reference

The M3 AI-use record is included in the single register in Section 16 and synchronized with registers/AI_Usage_Register.md.

Verification item	Check performed	Status
Staging slice	The project team reviewed the separate lifecycle-service slice and its 13/13 recorded checks.	Verified by the project team; 2026-10-16; slice scope retained

Verification item	Check performed	Status
Load profile	The project team reviewed the integrated website profile: 50 users, 6,000 measured samples, 647 failures (10.78333%), p95 1,051.639 ms.	Verified by the project team; 2026-10-17; reliability threshold failed
Integrated application	Reviewed M3 product, local test result and test links.	Verified by the project team for local M3 scope; 2026-10-16
CI	Checked successful main run 38870644636 on commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659.	Passed
Registers and decision	Reviewed the linked evidence and local-release recommendation.	Approved by team; 2026-10-16
Team M4 review	The project team reviewed the M4 document, calculations, release decision and retained evidence.	Approved collectively by the project team; 2026-10-17

43 Release Evidence Summary and recommendation

Recommendation: READY FOR THE M3 SCHOOL-PROJECT RELEASE AND LOCAL DEMONSTRATION.

The project team completed the required approvals and confirmed verification on 16 October 2026. The integrated M3 application is in the M3 product directory and is identified by main commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659. The recorded 37-test local suite passed, and GitHub Actions run 38870644636 passed on that main commit. The separate staging and load reports are identified as lifecycle-service evidence. The full summary is Release_Evidence_Summary.md.

Area	Evidence	Result	Limit
Application	Integrated M3 application source	Integrated citizen, staff and management local app present.	Local school-project use.
Local tests	M3 local test evidence	37 passed; 0 failed; 0 errors.	The team-reviewed local run reports the stated result; the recorded execution environment was Python 3.15.0.
CI	GitHub Actions run 38870644636 on main 8751bdf	Passed.	Applies to that exact commit.
Staging slice	M3 staging verification evidence	13/13 checks passed; team reran.	Separate lifecycle-service simulation.
Performance slice	M3 lifecycle-slice performance evidence	6,000 samples; zero failures; p95 0.825 ms.	Not integrated web-path performance.
Security	M3 web-security tests and Bandit/Gitleaks CI	Configured checks passed.	Team verification record for the local school-project scope.

Area	Evidence	Result	Limit
Planned M3 requirements and checks	Team verification record	Verified by the project team for the local school-project scope.	Team verification record for the local school-project scope.
Performance evidence	Integrated website profile: 6,000 samples; 647 failures (10.78333%); p95 1,051.639 ms.	Reliability threshold failed; production blocker. Separate lifecycle-slice performance is not a substitute.	The slice profile is not an integrated web-path measurement.
Team decision	Collective M4 team review approved 2026-10-17; production NO-GO.	Local academic demonstration only; protected-branch approvals and merge remain pending.	Scope is local.
AI usage	Unified project AI register; registers/AI_Usage_Register.md	The project team checked and verified every record; M3 approval 2026-10-16 and M4 collective approval 2026-10-17.	One project register is maintained in Section 16; GitHub review and merge remain separate repository controls.

43.1 Release scope and evidence

The M3 school-project release is ready to run locally. The project team confirmed that the planned requirements, tests and checks were tested, checked and verified, and completed the required approvals. Public launch and hosted production operations are outside the agreed scope.

The evidence identifies the local test run, the successful integrated main-branch CI run, the team-reviewed staging-slice results and their respective boundaries.

The project team reviewed the release evidence and current document set and completed the required approvals on 16 October 2026. M3 implementation PRs #13, #14 and #16 are merged.

Use README.md and the M3 product directory README.md to run the integrated M3 application locally with synthetic data. The separate staging runbook is for the lifecycle-service slice.

The team verified the planned M3 requirements and checks within the approved local school-project scope. Hosted availability and public production operations are outside that scope. Each evidence record retains its specific workload and boundary.

44 Evidence links and consistency check

Artefact	Purpose	Current record
CivicConnect_PED_v3.0.docx	Controlled M3 Project Engineering Document	Team-reviewed local-release decision
Release_Evidence_Summary.md	Release decision and evidence boundaries	M3 local school-project scope
registers/	Risk, defect and technical-debt registers; one AI register	Reviewed 2026-10-16

Artefact	Purpose	Current record
Integrated application evidence	M3 application test strategy and CI evidence	37 local tests; CI run 38870644636 on the integrated application commit.
evidence/	Staging, performance and robustness evidence	The project team reran and reviewed the linked evidence on 2026-10-16.
staging/	Separate from the integrated M3 application	Separate from integrated M3 app
staging/staging.config.json	0c71c5ff6730773fb0b51fccedaa27cd4fb00737ecbebf40d3678a11e960efce	Controlled staging configuration
staging/deploy_staging.py	3b0dead83e95300c1d228cbd39c056ccaacc4d3da1f9befcc20245d7c6f54c83	Clean-install deployment
staging/verify_staging.py	5dd4a55fe7a9d609670cc071d54fa5c9274e16cd9da3e601c6650086bb051795	13 staging checks
staging/load_test.py	8433f8f43a96a124427230e8c9ffc8ce3d7336a7c10611e795498eb944a3ab55	NFR-03 load test
staging/robustness_probe.py	8965790e91c11b8004c5fbec3e37d71f5ac5946bca1c4dcae2890239b3aa5db5	Malformed-input probe
evidence/staging_verification.json	86ffb2abdee325e913d04d9ba51bb394a5ccf3ef4c399ffce4443187bfcc3cb7	Staging results
evidence/perf_nfr03_profile.json	5c781f85275d0a3c55791421275fd32f2e86684fab496098b5dc19b3b8b4793d	Primary load results
evidence/perf_stress_supplementary.json	022bbb8f7c3ccc1574f50a7efef9449191be83fd15675b8f1b36983af86ede88	Headroom run results
evidence/robustness_probe.json	75cad5e72f7ce6e0eafd5870fc93da20d5d20ed4be5cb01590f83b8fb7ab148d	Probe results
docs/m4_final/ CivicConnect_PED_v4.0_Final.docx and .pdf	M4 final PED and matching rendered submission copy	Team review approved 2026-10-17; PDF regenerated from final DOCX
docs/m4_final/Final_RTM.csv and Scope_Completion.csv	Final traceability matrix and requirement-level scoring	20 signed-off M1 rows; paths checked; total 16.0/20 = 80.0%
docs/m4_final/ Final_Release_Decision.md	M4 release decision	Collectively approved 2026-10-17; production NO-GO
evidence/website-stage-20261016/	Integrated website staging and load evidence	Team-reviewed; 9/9 staging checks; load reliability gate failed
registers/AI_Usage_Register.md; Risk_Register_M3.md; Defect_Register.md; Technical_Debt_Register.md	Unified AI and current project risk, defect and debt registers	Team-reviewed; synchronized with M4 closure

M3 evidence review dated 16 October 2026: the integrated M3 application is identified by main commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659 and successful CI run 38870644636. The recorded local test suite reports 37 passing tests. The project team checked and verified the recorded M3 requirements, tests, checks and supporting evidence, and approved local school-project demonstration readiness on 16 October 2026. The integrated M4 website load profile subsequently failed the reliability threshold; the M4 production decision is NO-GO. Each evidence record retains its workload and scope boundary.

45. M4 Finalisation and Status Authority

This section records the final status of the same CivicConnect project. The M1 scope baseline is the signed-off CivicConnect_Milestone1_v0.11. Earlier M2 and M3 sections remain as project history. Where an earlier status conflicts with the evidence below, this M4 closure and the Final RTM take precedence.

Application and evidence baseline: main commit c8fcc9887d58bc07e7b592e292dbe05222d97190. Repository: https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup. The build is a local, server-rendered Python and SQLite application using synthetic demonstration data. No production release or hosted service is claimed. The project team completed and approved the collective M4 content and evidence review on 17 October 2026; supporting records were checked and verified. This document approval is separate from the required non-author GitHub pull-request reviews and merge, which remain pending.

46. M1 Scope to Final Delivery

The signed-off M1 baseline contains 20 requirement rows, each with one requirement-level acceptance-criteria entry: 13 functional and 7 non-functional requirements. The M4 scoring unit is one of these 20 signed-off rows; its complete acceptance-criteria cell is assessed as a single unit, including any clauses within that cell. Each row is scored once: Met = 1.0, Partially Met = 0.5, Not Met = 0.0. No new criterion IDs or post-sign-off scoring units have been introduced. The seven NFR rows remain in the denominator because they are part of the approved baseline.

Status group	Requirement IDs	Count	Points
Implemented and met	FR-01 to FR-13; NFR-02	14	14.0
Partially implemented / met	NFR-03, NFR-05, NFR-06, NFR-07	4	2.0
Not met	NFR-01, NFR-04	2	0.0
Total	20 signed-off M1 requirement rows	20	16.0 (80.0%)

Scope Completion Percentage = 16.0 ÷ 20 × 100 = 80%. The result meets the M4 target. NFR-01 is a Must Have criterion and remains unmet because the required five-person first-time-user study was not completed. The team deferred the study while prioritising correction of the load-reliability issue; there was insufficient time to complete it before M4 closure. The team reviewed and verified this status, and the omission reduces confidence in the first-time citizen submission experience. NFR-04 is unmet because the required 20-weekday availability probe is absent; current approved deployment is local demonstration, not hosted operation.

Scoring unit: one complete acceptance-criteria entry in each signed-off M1 requirement row. When an entry contains multiple clauses, the row is Met only when all material clauses are supported by evidence; it is Partially Met when at least one clause is supported and at least one remains incomplete; it is Not Met when no material clause is supported. This preserves the approved M1 structure and prevents post-sign-off splitting of criteria. The Final RTM retains the complete M1 text and one aggregate score per requirement row.

The detailed status, criterion text, implementation, evidence and limitation for all 20 entries are in docs/m4_final/Final_RTM.csv. The original signed-off M1 PED is retained at docs/CivicConnect_Milestone1_v0.11.docx.

47. Final Verification and M3 Issue Closure

The M3 team verification record reports 37 local regression tests passed and a successful GitHub Actions quality run on application commit 8751bdfabc4535a1aeb060e34d9fe2f5ce1d8659. These results support the tested functional and security paths only. The later integrated website profile adds materially different evidence and must be considered for M4 release readiness.

The website staging checks passed 9 of 9 listed checks, including authentication, stylesheet delivery, request list/detail rendering, cross-citizen denial, CSRF rejection, a citizen submission journey and responsiveness after load. The separate 10-minute load profile used 50 authenticated virtual users, 10,000 synthetic requests and 6,000 measured lookups. The p95 latency was 1,051.639 ms against a 3,000 ms limit. However, 647 of 6,000 samples failed (10.78333%) against the required failure rate below 1%. The load acceptance criterion therefore remains PARTIALLY MET and is a release blocker for production.

The connection resets occurred during synchronized request bursts against the single-process local WSGI server. The evidence identifies a possible pending-connection queue limit as a hypothesis, not a confirmed root cause. The cause remains open. The result applies to the tested local Windows loopback HTTP setup and does not establish public capacity, TLS behavior or production availability.

Issue / record	Final status	Evidence	Consequence / next action
NFR-03 load reliability	Partial; release blocker	evidence/website-stage-20261016/ staging_load_report.json and .md	Investigate connection resets; rerun the same 50-user, 10-minute profile. Require p95 <= 3000 ms and failures < 1% before production consideration.
NFR-01 usability	Unmet	The five-person study was deferred while the load-reliability issue was prioritised; insufficient time remained before M4 closure. The team reviewed and verified the reason and unmet status.	Complete the defined study or obtain a formally approved baseline change. Keep the Must Have unmet status visible.
NFR-04 availability	Unmet	No 20-weekday, one-minute probe evidence	Do not claim availability or production readiness. Define and collect the required probe evidence if hosted release enters scope.
DEF-001 to DEF-004	Prior dispositions retained; DEF-004 open as production release blocker	registers/Defect_Register.md; evidence/website-stage-20261016/staging_load_report.json	Diagnose the integrated website connection resets and repeat the load profile without weakening the M1 threshold.
Technical debt TD-004 to TD-014	Open or accepted per the current register; TD-012/013/014 remain M4	registers/Technical_Debt_Register.md	Retain the schema, local operations and runtime limitations, failed

	follow-up items		reliability gate, deferred usability study and absent hosted availability evidence.
AI usage register	All M1–M4 AI records are consolidated in the uniform project register in Section 16.	registers/AI_Usage_Register.md	The project team checked and verified each recorded AI-assisted activity; supporting evidence is retained in the project repository.

48. Final Deployment and Go-Live Decision

Final decision: NO-GO for production. The current evidence shows a 10.78333% HTTP request failure rate under the 50-user load profile, above the M1 limit of less than 1%, and no evidence for the 20-weekday availability target. Releasing to production on this evidence would not be justified.

The product may be demonstrated locally for academic assessment using synthetic accounts and data, with the local-only boundary stated before the demonstration. This does not constitute production or simulated-production readiness. No public URL, TLS deployment, operational monitoring, backup schedule or production rollback evidence is claimed.

Before production can be reconsidered, the team must identify and resolve the connection-reset cause, rerun the documented 50-user profile with the same acceptance thresholds, verify the final build identity, and supply the deployment, security, data-integrity and availability evidence required by the deployment target. The root cause and any corrective implementation remain open work.

49. Project Reflection and Conclusion

Original problem and stakeholder value

The local application provides a central citizen request workflow, requester status history, staff processing and management reporting. Recorded M3 evidence supports the exercised paths, while usability and hosted-operation evidence remain incomplete.

Scope and changes

The requirement-level completion score is 80%. The 13 functional requirement entries and NFR-02 are recorded as met. NFR-03, NFR-05, NFR-06 and NFR-07 are partial. NFR-01 and NFR-04 are unmet. Deferred advanced reporting, integrations, mobile application and advanced notifications remain outside the M1 committed scope.

Quality and uncertainty

The existing automated suite and CI provide evidence for recorded code paths. The project team checked and verified that the integrated website load profile met the p95 latency target but failed the request reliability gate, and that the required usability study and hosted availability records are absent. NFR-01 and NFR-04 remain unmet.

Engineering decisions and consequences

The local Python/SQLite architecture supported a low-cost integrated academic demonstration. The single-process local WSGI server did not satisfy the tested synchronized 50-user failure-rate gate. Keeping the deployment local protects users while reliability and operational concerns remain unresolved.

Next work

Investigate the connection resets, correct the serving configuration if required, repeat the load profile, complete or formally change the usability criterion, improve requirement-to-evidence closure, and collect availability evidence before any production decision.

50. Final Evidence Index and Submission Record

Submission item	Controlled location / value	Status
PED v4.0 Final	docs/m4_final/CivicConnect_PED_v4.0_Final.docx and CivicConnect_PED_v4.0_Final.pdf	Approved collectively by the project team; 17 October 2026. PDF regenerated from this DOCX.
Final RTM	docs/m4_final/Final_RTM.csv	Complete; 20 signed-off M1 requirement rows; full traceability and evidence paths checked.
M1 scope calculation	docs/m4_final/Scope_Completion.csv	Verified; 16.0/20 signed-off M1 requirement rows = 80.0%.
Final release decision	docs/m4_final/Final_Release_Decision.md	Approved collectively by the project team; production NO-GO; local synthetic-data demonstration only.
Team presentation	docs/m4_final/CivicConnect_Final_Demonstration.pptx	Prepared; collective review complete. Rehearsal remains a presentation-day activity.
Repository	https://github.com/VoidMeBro/SEN381-CivicConnect-Lineup	Public repository
Application/evidence baseline	c8fcc9887d58bc07e7b592e292dbe05222d97190	Exact main commit assessed in this closure
Final M4 documentation merge commit	Record the exact SHA after the protected-branch PR is approved and merged.	Pending required non-author GitHub reviews, merge and exact merge SHA.

Team final review	Collective M4 content and evidence approval; no individual decisions are recorded.	Approved collectively; 17 October 2026. Final content and retained evidence checked and verified.
Final submission date	Enter date at submission	Enter the actual date when the package is submitted.

The project team completed and approved the collective M4 content and evidence review on 17 October 2026. The team checked and verified the final document, calculations, release decision, evidence references and supporting records. This approval covers the document and project evidence. The separate GitHub control still requires the M4 documentation pull request to receive the required non-author reviews and merge before the final documentation merge SHA is recorded.

51. Version History

Version	Date	Change summary	Status
v1.0	2026-09-09	Signed-off M1 engineering foundation and requirements baseline, represented by PED v0.11.	Approved baseline
v2.0	2026-09-30	M2 architecture, design and lifecycle evolution.	Project history
v3.0	2026-10-16	Integrated M3 application, test evidence, registers and local demonstration release record.	M3 project record
v4.0	2026-10-17	M1 scope closure, final RTM, measured website load outcome, no-go decision and project conclusion.	Approved collectively by the project team for final submission; protected-branch reviews and merge pending.